

5G RAN

CA Experience Improvement Feature Parameter Description

Issue

Draft B

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added the impact relationship between downlink Inter-gNodeB CA and the downlink scheduling priority weight factor. For details, see 5.3.2.3 Function Impacts .	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added downlink RLC throughput distribution measurement for 3CC or higher CA UEs. For details, see: • 6.1 Principles • 6.4.1 Data Configuration • 8.1 Principles • 8.4.1 Data Configuration	Modified parameter: Added the value DL_NCC_UE to the NRDUCellOamParam.UeSpecStatisticsPolicy parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
 - Revised the statement that downlink fast intra-gNodeB CA is not supported in SUL cells. For details, see [7.3.4 Networking](#).
 - Added the dependency relationships of the maximum number of downlink carriers for CA with the switch for "SRS remote interference avoidance" and the switch for "SRS interference coordination based on self-contained slots". For details, see [8.3.2.1 Prerequisite Functions](#).
 - Revised descriptions of the benefits of uplink fast intra-gNodeB CA and the method (including the related counter) for activation verification. For details, see [7.2.1 Benefits](#) and [7.4.2 Activation Verification](#).
 - Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for the "downlink 8-layer transmission for a single FWA user" function by 3CC or higher CA UEs when the DL_EIGHT_LAYER_SCENARIO_EXT_SW option of the NRDUCellIDIRank.DIRankSelAlgoSw parameter is selected. For details, see 6.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a method for observing the activation and running state of 3CC aggregation. For details, see 6.4.2 Activation Verification .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added a method for observing the activation and running state of inter-gNodeB CA. For details, see 5.4.2 Activation Verification .	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added uplink fast intra-gNodeB CA. For details, see: 7.1 Principles 7.2.1 Benefits 7.3.2.1 Prerequisite Functions 7.4.1.1 Data Preparation 7.4.1.2 Using MML Commands	Modified parameter: Added the UL_CA_FAST_SPLIT_SW option to the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the "adaptive HARQ resource limitation for CA" function. For details, see: 5.1.3 Functions for Limiting HARQ Resources for CA 5.2.1 Benefits 5.2.2 Impacts 5.3.2.1 Prerequisite Functions 5.4.1.1 Data Preparation 5.4.1.2 Using MML Commands	Modified parameter: Added the CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW option to the NRDUCellPucch.PucchAlgoExtSwitch parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added descriptions of the "uplink enhancement for smart carrier selection" function. For details, see: • 6.2.2 Impacts • 8.2.2 Impacts For details about the principles of this function, see <i>Carrier Aggregation</i>.	Modified parameter: Added the SMART_CARR_SEL_UL_ENH_SW option to the NRDUCellCollabServ.MultiCarrMgmtSw parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added descriptions of the "reselection policy for 3CC and massive CA UEs in idle mode" function. For details, see: • 6.1 Principles, 6.2.1 Benefits, 6.4.1.1 Data Preparation, and 6.4.1.2 Using MML Commands • 8.1 Principles, 8.2.1 Benefits, 8.4.1.1 Data Preparation, and 8.4.1.2 Using MML Commands	Added the NRCelOpQciBearer.ServiceDiffAlgoSwitch parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added the impact relationships of the reselection policy for 3CC and massive CA UEs in idle mode with the T320 timer and the low-frequency dedicated-priority-based idle mode MLB function. For details, see 6.3.2.3 Function Impacts.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Moved the description of the dependency relationship of SRS carrier switching enhancement and 3CC aggregation /massive CA from [4.3.2.3 Function Impacts](#) and [6.3.2.3 Function Impacts](#) to [4.3.2.1 Prerequisite Functions](#).

Moved the description of the dependency relationship between inter-gNodeB CA and PUCCH RB adaptation from [5.3.2.3 Function Impacts](#) to [5.3.2.1 Prerequisite Functions](#).

Moved descriptions of the **CA_HARQ_RES_LIMIT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter from [5.2.2 Impacts](#) to [5.1.3 Functions for Limiting HARQ Resources for CA](#).

Changed the description of the RF modules required for CA-UE-specific frequency-priority-based inter-frequency handover from "All NR-capable RF modules support this function" to "All NR-capable RF modules that work in low frequency bands support this function". For details, see [9.3.3 Hardware](#).

Added a note on Xn link setups for inter-gNodeB inter-FR CA. For details, see [5.3.4 Networking](#).

Revised the impact relationship between 3CC aggregation and downlink 8-layer transmission for a single FWA user. For details, see [6.3.2.3 Function Impacts](#).

Renamed "fast intra-gNodeB CA" and "fast inter-gNodeB CA" as "downlink fast intra-gNodeB CA" and "downlink fast inter-gNodeB CA", respectively. For details, see [7 Fast CA](#).

Added the dependency relationships of the maximum number of downlink carriers for CA with the switch for "SRS remote interference avoidance" and the switch for "SRS interference coordination based on self-contained slots". For details, see [6.3.2.1 Prerequisite Functions](#).

Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-050206	CA SRS Carrier Switching	4 CA SRS Carrier Switching
FOFD-051202	Inter-gNodeB CA	5 Inter-gNodeB CA
FOFD-051211	3CC CA	6 3CC Aggregation
FOFD-061224	Fast CA	7 Fast CA
FOFD-080012	Massive CA	8 Massive CA

 NOTE

Massive CA is a commercial feature in a general sense. However, it is for trial use in certain scenarios. Specifically:

- NR TDD massive CA is available for trial use.

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
CA SRS carrier switching	None	4 CA SRS Carrier Switching
Inter-gNodeB CA	None	5 Inter-gNodeB CA
3CC aggregation	None	6 3CC Aggregation
Fast CA	None	7 Fast CA
Massive CA	Supported only in SA networking	8 Massive CA
CA-UE-specific frequency-priority-based inter-frequency handover	None	9 CA-UE-specific Frequency-Priority-based Inter-Frequency Handover

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
CA SRS carrier switching	Supported only in low frequency bands	4 CA SRS Carrier Switching
Inter-gNodeB CA	Supported in both low and high frequency bands, with the following major differences: In low-frequency scenarios, inter-gNodeB CA requires the PCell and SCells all to be low-frequency cells. In inter-FR CA scenarios, inter-gNodeB CA works only between a low-frequency PCell and high-frequency SCells. These SCells must be configured on the same baseband processing unit.	5 Inter-gNodeB CA
3CC aggregation	Supported only in low frequency bands	6 3CC Aggregation
Fast CA	Supported only in low frequency bands	7 Fast CA
Massive CA	Supported only in low frequency bands	8 Massive CA
CA-UE-specific frequency-priority-based inter-frequency handover	Supported only in low frequency bands	9 CA-UE-specific Frequency-Priority-based Inter-Frequency Handover

3 Overview

3.1 Definition

3GPP Release 15 requires that individual New Radio (NR) UEs work with up to 1 GHz of bandwidth so that users can enjoy services at higher rates. However, operators' spectrum resources may not be contiguous or may exceed the single-carrier bandwidth capability defined in 3GPP specifications. Against this backdrop, 3GPP introduced carrier aggregation (CA), which allows for aggregation of multiple contiguous or non-contiguous component carriers (CCs) to expand bandwidth to a 3GPP-required value. This document describes the functions related to CA experience improvement. For details about the basic principles and scenarios of CA, see *Carrier Aggregation*.

3.2 Overview of CA Functions

gNodeBs offer various CA functions to CA UEs according to UE-reported CA capabilities and the carrier management mechanisms described in *Carrier Aggregation*. These functions allow CA UEs to have larger bandwidths for data transmission. [Table 3-1](#) and [Table 3-2](#) provide an overview of these functions.

Table 3-1 Overview of CA functions (1)

Function Name	Intra-Band	Inter-Band	Intra-gNodeB	Inter-gNodeB	Uplink	Downlink
Downlink intra-gNodeB intra-band CA	√	-	√	-	-	√
Uplink intra-gNodeB intra-band CA	√	-	√	-	√	-

Function Name	Intra-Band	Inter-Band	Intra-gNodeB	Inter-gNodeB	Uplink	Downlink
Flexible carrier management (high-frequency TDD)	√	-	√	-	√	√
Downlink intra-gNodeB intra-FR inter-band CA	-	√	√	-	-	√
Uplink intra-gNodeB intra-FR inter-band CA	-	√	√	-	√	-
Inter-FR inter-band CA	-	√	-	√	√	√
CA SRS carrier switching	√	√	√	√	-	√
Inter-gNodeB CA	√	√	-	√	-	√
3CC aggregation	√	√	√	√	-	√
Fast CA	√	√	√	√	-	√
Massive CA	√	√	√	√	-	√
CA-UE-specific frequency-priority-based inter-frequency handover ^a	√	√	√	√	√	√

Function Name	Intra-Band	Inter-Band	Intra-gNodeB	Inter-gNodeB	Uplink	Downlink
a: CA-UE-specific frequency-priority-based inter-frequency handover is an inter-frequency handover procedure specific to CA UEs that is performed to change the PCell for each CA UE. This function works in all the scenarios listed in this table.						

 NOTE

In the preceding table, "√" indicates that the function is supported in the specified scenario, while "-" indicates that the function is not supported.

Table 3-2 Overview of CA functions (2)

Function Name	Function Switch	Reference
Downlink intra-gNodeB intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>
Uplink intra-gNodeB intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>
Flexible carrier management (high-frequency TDD)	NRDUCellCarrMgmt.Fr2CaCarrMgmtPolicy set to USER_TRAFFIC_BASED	<i>Carrier Aggregation</i>
Downlink intra-gNodeB intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>
Uplink intra-gNodeB intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>
Inter-FR inter-band CA	INTER_FR_FR1TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>
CA SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	4 CA SRS Carrier Switching

Function Name	Function Switch	Reference
Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	5 Inter-gNodeB CA
3CC aggregation	NRDUCellCarrMgmt.CaDLMaxCcNum set to DL3CC	6 3CC Aggregation
Fast CA	Fast intra-gNodeB CA: CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter Fast inter-gNodeB CA: INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	7 Fast CA
Massive CA	4CC aggregation: NRDUCellCarrMgmt.CaDLMaxCcNum set to DL4CC 5CC aggregation: NRDUCellCarrMgmt.CaDLMaxCcNum set to DL5CC	8 Massive CA
CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRI_BASED_IN TERF_HO_SW option of the NRCCellAlgoSwitch.InterFreqHoSwitch parameter	9 CA-UE-specific Frequency-Priority- based Inter-Frequency Handover

4 CA SRS Carrier Switching

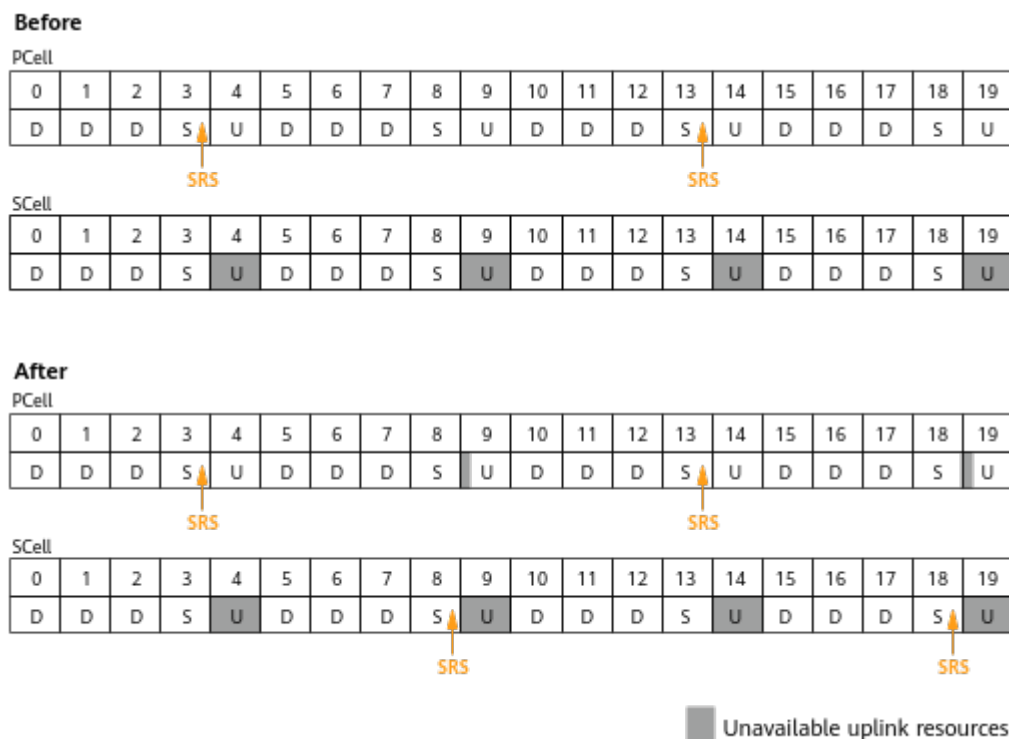
4.1 Principles

There are two types of SCCs: uplink+downlink SCCs and downlink-only SCCs. SRS measurements are available in SCells on uplink+downlink SCCs (uplink+downlink SCells). Therefore, these SCells have the same downlink beamforming performance as PCells. In contrast, SCells on downlink-only SCCs (downlink-only SCells) lack SRS-based weights and have to use PMI-based weights. This compromises downlink beamforming performance in TDD downlink-only SCells. SRS carrier switching can be enabled for PCells to help TDD downlink-only SCells obtain SRS resources and improve the downlink beamforming performance in these cells.

SRS Carrier Switching

With SRS carrier switching enabled, a UE can send SRSs in its PCell and an SCell in TDM mode. After receiving the SRSs, the gNodeB evaluates the downlink channel quality based on the SRSs and calculates SRS-based weights for beamforming in the SCell. [Figure 4-1](#) illustrates how SRS carrier switching works. Note that, after SRS carrier switching is enabled, certain symbols in uplink slots of the PCell will become unavailable. That is because there is delay in uplink transmission switching from the SCell to the PCell.

Figure 4-1 Principles for SRS carrier switching



SRS carrier switching is controlled by the **SRS_CARRIER_SWITCHING_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter. In NSA networking, this function also requires that the **SRS_CARR_SWITCH_CAPB_QUERY_SW** option of the LTE parameter **NsaDcAlgoParam.NsaDcAlgoSwitch** be selected.

As SRS carrier switching allows SRSs to be transmitted in the PCell and an SCell in turn, a short SRS period in the PCell needs to be extended to leave opportunities for SRS transmission in the SCell. The occasions where such an extension is needed can be automatically identified by the gNodeB if the **CARR_AGG_SRS_ALLOC_SW** option of the **NRDUCellSrs.SrsAlgoExtSwitch** parameter is selected. With this option selected, the gNodeB can also set an appropriate SRS period for the PCell.

The slot configuration combinations compatible with SRS carrier switching (2CC) are the same as those for intra-FR inter-band CA, which are described in *Carrier Aggregation*.

SRS Carrier Switching Enhancement

SRS carrier switching enhancement allows two TDD SCells to participate in SRS carrier switching at the same time. It works in 3CC aggregation scenarios with a TDD cell acting as the PCell and two TDD cells acting as SCells. This function is controlled by the **SRS_CARRIER_SWITCHING_ENH_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter and requires SRS carrier switching to be enabled.

The slot configuration combinations compatible with SRS carrier switching enhancement are described in *Carrier Aggregation*.

4.2 Network Analysis

4.2.1 Benefits

The gains of CA SRS carrier switching depend on the radio propagation condition (LOS or NLOS):

- When SRS carrier switching takes effect in line of sight (LOS) scenarios, using SRS-based weights in SCells delivers only limited gains over using PMI-based weights. In addition, if the SRS frequency hopping multiplier is large and therefore wideband SRS measurement takes a long time (such as 680 ms), using SRS-based weights in SCells delivers limited or even no gains over using PMI-based weights.
- When SRS carrier switching takes effect in non-line-of-sight (NLOS) scenarios:
 - SRS-based weights can be used for beamforming in TDD SCells. This increases the downlink throughput of UEs in these SCells by an average of 20% to 30%. A shorter SRS transmission interval in the SCells leads to a higher gain, and a longer interval leads to a smaller gain.
 - A UE will transmit SRSs in its PCell and SCell in turn. If the SRS transmission interval in the PCell is extended as a result, then:
 - If the UE is stationary or moving at a low speed, the extension has a slight impact on the downlink throughput of the UE in the PCell, with a throughput decrease of less than 5%.
 - If the UE is moving faster than 30 km/h, the extension will cause the downlink throughput of the UE in the PCell to decrease significantly. The decrease will partially offset the downlink throughput gain of the UE in the SCell. As a result, the downlink throughput gain of the UE decreases.

The longer the SRS transmission interval in the PCell, the larger the decrease in the downlink throughput of the UE in the PCell and the larger the decrease in the downlink throughput gain of the UE. In this situation, it is recommended that UE-characteristic-based SRS period adaptation be enabled (by selecting the **USER_CHARACTER_SRS_ADAPT_SW** option of the **NRDUCellSrs.SrsAlgoSwitch** parameter) together with SRS carrier switching so that SRS resources can be obtained in the PCell and SCell at a shorter interval. This way, SRS carrier switching provides more significant gains. For details about UE-characteristic-based SRS period adaptation, see *Channel Management*.

- After SRS carrier switching enhancement is enabled, SRS-based weights can be used in both SCells, and the downlink throughput of CA UEs is expected to be 1% to 5% higher than that achieved using SRS carrier switching.

For an NR TDD SCell served by a LampSite base station, the gains brought by switching from PMI-based weights to SRS-based weights are available only when distributed massive MIMO is also enabled (by selecting the **DM_MIMO_SERVICE_SWITCH** option of the **NRDUCellAlgoSwitch.DmMimoSwitch** parameter).

After SRS carrier switching is enabled, the average throughput of SCell-activated CA UEs and that of SCell-configured CA UEs may not increase in scenarios with a relatively large number of small-packet services. That is because the protocol-defined method of calculating these throughputs does not count in tail packets, but SRS carrier switching boosts CA UE scheduling in their SCells and increases the proportion of tail packets in these scenarios. The throughputs are calculated as follows:

- Average throughput of SCell-activated CA UEs =
$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)}{N.CA.ThpTime.DL.RmvLastSlot}$$
- Average throughput of SCell-configured CA UEs =
$$\frac{(N.CA.SccConfigUser.ThpVol.DL - N.CA.SccConfigUser.ThpVol.DL.LastSlot)}{N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot}$$

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

- After SRS carrier switching is enabled, the value of *switchingTimeUL* in the *SRS-SwitchingNR* IE reported by most UEs is not 0. This results in inconsistent PUSCH symbol lengths between SRS carrier switching UEs and common UEs, which in turn affects the uplink performance of UEs. To mitigate this impact, the gNodeB needs to support scheduling of multiple symbol lengths in the same slot, which is known as scheduling of multi-SLIV UEs (SLIV is short for start and length indicator value). This scheduling function is controlled by the **NRDUCellPusch.SchMultiSlivTypeUeSwitch** parameter. It must be enabled before SRS carrier switching is enabled.
- After SCells are activated for a UE, uplink services are interrupted when the UE sends SRSs in the PCell and SCells in turn, affecting the uplink throughput of the UE in the PCell. The uplink throughput is no longer affected once the SCells are deactivated.
- In heavy-load scenarios, the values of the following counters increase:
 - **N.RRC.ReEst.Succ.Total**, which measures the total number of successful RRC reestablishments with a cell
 - **N.RRC.ReEst.Att.Total**, which measures the total number of requests for RRC reestablishments with a cell
 - **N.CA.SCell.Deact.Att**, which measures the number of SCell deactivation executions for CA UEs
 - **N.CA.SCell.Deact.Succ**, which measures the number of successful SCell deactivations for CA UEs
 - **N.CA.SCell.Rmv.Att**, which measures the number of SCell removal attempts for CA UEs
 - **N.CA.SCell.Rmv.Succ**, which measures the number of successful SCell removals for CA UEs
 - **N.CA.SCell.Add.Att**, which measures the number of SCell addition attempts for CA UEs
 - **N.CA.SCell.Add.Succ**, which measures the number of successful SCell additions for CA UEs

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-050206	CA SRS Carrier Switching	NR0S00CAWW00	per Cell
For the license control item NR0S00CAWW00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation: – When the license sales number pre-check function is enabled, the SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter cannot be modified due to license insufficiency, preventing cell activation failures. – When the license sales number pre-check function is disabled, the cell cannot be activated again due to license insufficiency after the feature is enabled and the parameters that cause automatic cell reset are modified. The license sales number pre-check function is controlled by the LIC_SALE_NUM_PRE_CHECK_SW option of the gNBLicenseAlmThld.LicenseChkSw parameter.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	SRS carrier switching (controlled by the SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that intra-band CA be enabled.
Low-frequency TDD	Intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	SRS carrier switching (controlled by the SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that intra-FR inter-band CA be enabled.
Low-frequency TDD	SRS carrier switching	SRS_CARRIER_SWITCHING_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter	<i>CA Experience Improvement</i>	SRS carrier switching enhancement (controlled by the SRS_CARRIER_SWITCHING_ENH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that SRS carrier switching be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	3CC aggregation and massive CA	NRDUCellCarrMgmt.CaDLMaxCcNum	<i>CA Experience Improvement</i>	SRS carrier switching enhancement (controlled by the SRS_CARRIER_SWITCHING_ENH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that the NRDUCellCarrMgmt.CaDLMaxCcNum parameter be set to DL3CC/DL4CC/DL5CC .

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TDM power control	NSA_DC_ENH_UL_POWER_CONTROL_SW option of the NRCellNsaDcConfig.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Performance Enhancement</i>	CA SRS carrier switching does not work with TDM power control. They cannot be both enabled.
Low-frequency TDD	Secondary intermodulation interference avoidance	CROSS_MDLT_INTRF_AVOID_SW option of the NRDUCellAlgoSwitch.NsaDcAlgoSwitch parameter	<i>EPC-based NSA Basics</i>	CA SRS carrier switching does not work with secondary intermodulation interference avoidance. They cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Channel Management</i>	SRS interference coordination based on self-contained and uplink slots can be enabled only if CA SRS carrier switching is disabled.
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode and NRDUCellMultiTRP.DataTransMode	<i>Cell Combination</i>	CA SRS carrier switching cannot be enabled in a cell where Joint Transmission Cell Combination has been enabled (by setting NRDUCell.NrDuCellNetworkingMode to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode to JOINT_MODE .)

4.3.2.3 Function Impacts

- Impact relationships with basic functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SR period adaptation	SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResoureAlgoSwitch parameter	<i>Channel Management</i>	- SRS carrier switching does not take effect when the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResoureAlgoSwitch parameter is deselected and the NRDUCellPucch.SrPeriod parameter is set to a value less than SLOT20. - SRS carrier switching can take effect and does not impact the UEs' adaptively configured SR period when SR period adaptation (specified by the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResoureAlgoSwitch parameter) is disabled and the NRDUCellPucch.SrPeriod parameter is set to a value greater than or equal to SLOT20. - SRS carrier switching can take effect when the SR_PERIOD_ADAPT_SWITCH option of the NRDUCellPucch.SrResoureAlgoSwitch parameter is selected. In this case, the adaptively

RAT	Function Name	Function Switch	Reference	Description
				configured SR period is at least 20 slots for UEs on which SRS carrier switching takes effect.
Low-frequency TDD	UE-characteristic-based SRS period adaptation	USER_CHARACTERISTICS_SRS_ADAPT_SW option of the NRDUCells parameter <i>SrsAlgoSwitch</i>	<i>Channel Management</i>	After UE-characteristic-based SRS period adaptation is enabled for a cell, UEs in the cell can be allocated SRS resources at a shorter interval, increasing the gains of SRS carrier switching.
Low-frequency TDD	Rate-matching-pattern-configuration-based PDCCH rate matching	PDCCH_RATEMATCH_SW option of the NRDUCellPdsch parameter <i>RateMatchSwitch</i>	<i>Downlink Scheduling</i>	When CA SRS carrier switching is enabled, the increase in downlink throughput brought by rate-matching-pattern-configuration-based PDCCH rate matching slightly decreases.
Low-frequency TDD	Rate-matching-pattern-configuration-free PDCCH rate matching	PDCCH_NO_PATTERN_RATEMATCH_SW option of the NRDUCellPdsch parameter <i>PdcchAlgoSwitch</i>	<i>Downlink Scheduling</i>	When CA SRS carrier switching is enabled, the probability of rate-matching-pattern-configuration-free PDCCH rate matching taking effect decreases and the increase in downlink throughput slightly decreases.

- Impact relationships with functions related to carrier management

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	32-port CSI-RS measurement on SCCs	CA_SCC_CSIRS_32PORT_SW option of the NRDUCells.CsiSwitch parameter	<i>Carrier Aggregation</i>	32-port CSI-RS measurement on SCCs does not take effect for CA UEs for which SRS carrier switching is in effect.
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When uplink intra-band CA is enabled together with SRS carrier switching or SRS carrier switching enhancement, only one of them can take effect at a time. Which of them takes effect is dependent on the multi-carrier capability combination selection policy, which is specified by the NRCellCaMgmtConfig.CaBandCombSelectionStrat parameter.
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	When uplink intra-FR inter-band CA is enabled together with SRS carrier switching or SRS carrier switching enhancement, only one of them can take effect at a time. Which of them takes effect is dependent on the multi-carrier capability combination selection policy, which is specified by the NRCellCaMgmtConfig.CaBandCombSelectionStrat parameter.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic-codebook-based HARQ feedback over the PUCCH	PUCCH_DYNAMIC_HARQ_ACK_CB_SWITCH option of the NRDUCellCarrierMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Carrier Aggregation</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for UEs for which SRS carrier switching has taken effect and from which <i>switchingTimeUL</i> with a value other than 0 μ s is reported.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SWITCH option of the NRDUCellCarrierMgmt.CaEnhancedAlgoSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When SRS carrier switching takes effect in a UE's SCell that is involved in joint scheduling, SRS power control adjustment does not take effect for the UE in this SCell. For a small number of UEs that cannot be scheduled in the downlink N symbols before SRS transmission, the PDSCH may not be scheduled in self-contained slots of the scheduling-dominant cell because the number of start and length indicator value (SLIV) combinations is not enough.

- Impact relationships with other functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink DMRS adaptation	VOICE_DMRS_ADAPT_SW option of the NRDUCellDmrs.UIDmrsSwitch parameter	<i>VoNR</i>	When CA SRS carrier switching takes effect for UEs with VoNR enabled, uplink DMRS adaptation does not take effect in an uplink slot in which the PUSCH occupies at most 12 symbols.
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	Search space set group (SSSG) switching does not take effect for CA SCell-activated UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Positioning-dedicated SRS transmission in uplink slots	SRS_FOR_POSITION_ON_USLOT_SW option of the NRDUCellPosAlgo.LcsMeasureAlgorithmSw parameter	<i>Positioning</i>	When both positioning-dedicated SRS transmission in uplink slots and CA SRS carrier switching are enabled: • The setting of the SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrs.SrsAlgoExtSwitch parameter must be the same between the PCell and SCells. Otherwise, CA SRS carrier switching does not take effect. • If PUCCH format 3 resources are insufficient, the User Downlink Average Throughput (DU) slightly decreases.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_ID_SW and SRS_INTRF_COORD_SLOT_SW options of the NRDUCells parameter <i>SrsAlgoExtSwitch</i>	<i>Remote Interference Management (Low-Frequency TDD)</i>	- In a cell with proactive avoidance of remote interference enabled, the CA SRS carrier switching switch can be turned on only if the NRDUCellCarrMgmt.SrsCarrSwitchingULSymCfg and NRDUCellCarrMgmt.SrsCarrSwitchingTimeULThld parameters are set to the same value. - When proactive avoidance of remote interference is enabled, CA SRS carrier switching or CA SRS carrier switching enhancement works only in certain scenarios^a. - If proactive avoidance of remote interference is enabled together with CA SRS carrier switching or CA SRS carrier switching enhancement in a cell, the following network impacts arise: - CA SRS carrier switching takes effect for a smaller number of times for UEs that treat the

RAT	Function Name	Function Switch	Reference	Description
				cell as their SCell. As a result, the number of UEs in the CA state in the cell decreases. - The average uplink cell and UE throughputs may decrease.
a: When proactive avoidance of remote interference is enabled, CA SRS carrier switching or CA SRS carrier switching enhancement works only in the following scenarios: • A UE treats a TDD cell as its PCell and has one TDD SCell configured, the PCell and SCell use the same slot configuration, and proactive avoidance of remote interference is enabled in both of the cells. In this scenario, CA SRS carrier switching can take effect for the UE. • A UE treats a TDD cell as its PCell and has two TDD SCells configured, the PCell and SCells use the same slot configuration, proactive avoidance of remote interference is enabled in the PCell, and: - Proactive avoidance of remote interference is enabled in only one of the SCells. In this scenario, CA SRS carrier switching but not CA SRS carrier switching enhancement can take effect for the UE. - Proactive avoidance of remote interference is enabled in both of the SCells. In this scenario, CA SRS carrier switching enhancement can take effect for the UE.				

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

For PCells: All RRUs, pRRUs, and eCPRI-capable AAUs support this function.

For SCells:

- If a UBBPg/UBBPh is used, all RRUs, pRRUs, and AAUs support this function.
- If a UBBPfw is used, eCPRI-capable AAUs support this function. The TX/RX mode of cells must be 32T32R or higher.

For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Networking

According to section 4.2 "Transmission timing adjustments" in 3GPP TS 38.213 V17.3.0, the TA offset values for the PCell and SCells in the same TA group must be the same. If the TA offset values (specified by the **NRDUCell.TaOffset** parameter) for the cells involved in CA are different, SRS carrier switching cannot be configured.

4.3.5 Others

SRS carrier switching has the following requirements for UEs:

- UEs must support SRS carrier switching. That is, the UE-reported capability information must include the *SRS-SwitchingTimeNR* IE.
- UE must support antenna switching. That is, the UE-reported capability information must include the *SupportedSRS-TxPortSwitch* IE.
- The UE-reported capability information must include the SRS resource capability in the target band of SRS carrier switching, that is, the *supportedSRS-Resources* IE in *FeatureSetDownlink-v15a0*.
- The value of *switchingTimeDL* reported by UEs is less than or equal to the value of the **NRDUCellCarrMgmt.SrsCarrSwitchingTimeDLThld** parameter.
- The value of *switchingTimeUL* reported by UEs is less than or equal to the value of the **NRDUCellCarrMgmt.SrsCarrSwitchingTimeULThld** parameter.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-1 describes the parameters used for function activation. No parameters are involved in function optimization.

Before enabling SRS carrier switching for intra-band CA, enable downlink intra-band CA by preparing data as described in *Carrier Aggregation*.

Before enabling SRS carrier switching for intra-FR inter-band CA, enable downlink intra-FR inter-band CA by preparing data as described in *Carrier Aggregation*.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	SRS_CARRIER_SWITCHING_SW	You are advised to select this option if there are UEs that support SRS carrier switching on the network.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	SRS_CARRIER_SWITCHING_ENH_SW	You are advised to select this option if there are UEs that support 3CC aggregation and SRS carrier switching enhancement on the network.
SRS Carrier Switching Time DL Threshold	NRDUCellCar rMgmt.SrsCa rrSwitchingTi meDLThld	N/A	The value N0 is recommended.
SRS Carrier Switching Time UL Threshold	NRDUCellCar rMgmt.SrsCa rrSwitchingTi meULThld	N/A	The value N140 is recommended.
SRS Carrier Switching UL Symbol Config	NRDUCellCar rMgmt.SrsCa rrSwitchingU LSymCfg	N/A	The value N30 is recommended.
NSA DC Algorithm Switch	NsaDcAlgoPa ram.NsaDcAl goSwitch	SRS_CARR_SWITCH_CAPB_QUERY_SW	This parameter is set on the LTE side. You are advised to select this option if SRS carrier switching is to be enabled in NSA networking. Deselect this option if SRS carrier switching is to be enabled in SA networking.
SRS Algorithm Extension Switch	NRDUCellSrs. SrsAlgoExtSw itch	CARR_AGG_SRS_ALLOC_SW	You are advised to select this option for low-frequency TDD cells and deselect this option for other cells.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling SRS carrier switching in NSA networking
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=SRS_CARR_SWITCH_CAPB_QUERY_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_SW-1;
//Enabling SRS carrier switching in SA networking
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_SW-1;
//Enabling SRS carrier switching enhancement in NSA networking
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=SRS_CARR_SWITCH_CAPB_QUERY_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_ENH_SW-1;
//Enabling SRS carrier switching enhancement in SA networking
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_ENH_SW-1;
//Turning on the switch for prolonging SRS periods for CA UEs (If this switch is turned off, SCells may have
no SRS resources.)
MOD NRDUCELLSRS: NrDuCellId=0, SrsAlgoExtSwitch=CARR_AGG_SRS_ALLOC_SW-1;
//((Optional) Setting the SrsCarrSwitchingTimeDlThld parameter
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SrsCarrSwitchingTimeDlThld=N0;
//((Optional) Setting the SrsCarrSwitchingTimeUlThld parameter
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SrsCarrSwitchingTimeUlThld=N140;
//((Optional) Setting the SrsCarrSwitchingUlSymCfg parameter
MOD NRDUCELLCARRMGMT: NrDuCellId=0, SrsCarrSwitchingUlSymCfg=N30;
```

Deactivation Command Examples

```
//Disabling SRS carrier switching in NSA networking
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_SW-0;
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=SRS_CARR_SWITCH_CAPB_QUERY_SW-0;
//Disabling SRS carrier switching in SA networking
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_SW-0;
//Disabling SRS carrier switching enhancement in NSA networking
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_ENH_SW-0;
MOD NSADCALGOPARAM: NsaDcAlgoSwitch=SRS_CARR_SWITCH_CAPB_QUERY_SW-0;
//Disabling SRS carrier switching enhancement in SA networking
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=SRS_CARRIER_SWITCHING_ENH_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

- Step 1** Start Uu signaling tracing on the MAE as follows: Log in to the MAE and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > Uu Interface Trace**.
- Step 2** Enable a UE to access the network and check the fields in the *SCellConfig* IE in the RRCReconfiguration message delivered by the gNodeB during SCell configuration. If the IE contains the *carrierSwitching* field, this function has taken effect.

----End

4.4.3 Network Monitoring

Observe the average downlink UE throughput and average uplink UE throughput to check whether the UE data rate increases after this function is enabled.

- Average downlink UE throughput = **User Downlink Average Throughput (DU)**
- Average uplink UE throughput = **User Uplink Average Throughput (DU)**

5 Inter-gNodeB CA

5.1 Principles

Inter-gNodeB CA aggregates carriers that belong to different gNodeBs, between which data is transmitted over the eXn interface. Inter-gNodeB CA works only in the downlink. Downlink inter-gNodeB CA is controlled by the **INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter.

The following describes the overall procedure of inter-gNodeB CA, including its differences from intra-gNodeB CA:

1. During initial access, an incoming RRC reestablishment, or an incoming handover of a CA UE to its PCell, the UE reports its CA capabilities to the target gNodeB after its RRC connection is set up.
2. The gNodeB selects appropriate inter-gNodeB cells as SCells for the UE. The procedures for SCell configuration, change, activation, deactivation, and removal are similar to those described in *Carrier Aggregation*, with the following differences:
 - SCell configuration: Inter-gNodeB CA allows for measurement-based SCell configuration but not blind SCell configuration. Inter-gNodeB CA has candidate cell requirements, described in [5.1.1 Candidate SCell Requirements of Inter-gNodeB CA](#), in addition to those of intra-gNodeB CA.
 - SCell deactivation and removal: The eXn congestion status and the quality and delay of transmission between gNodeBs affect the SCell deactivation and removal procedures. Specifically:
 - If the eXn interface is congested but the transmission delay does not exceed the value of the **NRDUCellCarrMgmt.InterGnodebDelay** parameter, the gNodeB deactivates the SCells involved in inter-gNodeB CA.
 - If inter-gNodeB transmission quality deteriorates (for example, the transmission delay exceeds the value of the **NRDUCellCarrMgmt.InterGnodebDelay** parameter or the transmission is interrupted), the gNodeB removes the SCells to exit

inter-gNodeB CA. At least 10 minutes after SCell removal, the gNodeB can configure SCells again if transmission quality has recovered.

3. After activating the SCells, the serving gNodeB of the PCell distributes downlink data of the UE from the PCell to the inter-gNodeB SCells. For details about inter-gNodeB data distribution, see [5.1.2 Inter-gNodeB Data Distribution for CA](#).

NOTE

Uplink data is transmitted in the PCell.

To prevent inter-gNodeB CA UEs from affecting the HARQ resource request success rate of other UEs, functions for limiting HARQ resources for CA can be used. For details, see [5.1.3 Functions for Limiting HARQ Resources for CA](#).

For the band+bandwidth combinations compatible with inter-gNodeB CA, see *Carrier Aggregation*. Inter-gNodeB CA works in high and low frequency bands with the following constraints applied:

In low-frequency scenarios, inter-gNodeB CA requires the PCell and SCells all to be low-frequency cells.

In inter-FR CA scenarios, inter-gNodeB CA works only between a low-frequency PCell and high-frequency SCells. These SCells must be configured on the same baseband processing unit. For more details about inter-FR CA, see *Carrier Aggregation*.

5.1.1 Candidate SCell Requirements of Inter-gNodeB CA

Inter-gNodeB CA has the following requirements for candidate SCells:

- Only cells that are configured as neighboring cells for inter-gNodeB CA can be candidate SCells for downlink inter-gNodeB CA. For details about such neighboring cells, see [5.1.1.1 Candidate SCell Requirement on the Inter-gNodeB Flag for Neighboring Cells](#).
- The delay between inter-gNodeB cells must meet certain requirements described in [5.1.1.2 Candidate SCell Requirement on Delay](#).

5.1.1.1 Candidate SCell Requirement on the Inter-gNodeB Flag for Neighboring Cells

Only cells with the **NRCellRelation.InterGnodebFlag** parameter set to **TRUE** can act as candidate SCells for downlink inter-gNodeB CA. This parameter is used to configure the inter-gNodeB flag for neighboring cells to be involved in CA. It can be set in either of the following modes:

- Manual mode: If the **NRCellAnr.InterGnbCellCfgMode** parameter is set to **MANUAL**, the inter-gNodeB flag can only be manually set.
- Automatic mode: If the **NRCellAnr.InterGnbCellCfgMode** parameter is set to **AUTO**, the inter-gNodeB flag can be set both manually and automatically. When an inter-gNodeB cell meets the relevant threshold for event A5, the gNodeB automatically sets the flag to **TRUE** for the cell. If a cell is not configured as an SCell within seven days since the inter-gNodeB flag is set to **TRUE** for the cell, the gNodeB automatically sets the inter-gNodeB flag to **FALSE** for the cell.

 NOTE

In automatic mode, a gNodeB cannot predict whether there are cells that can act as candidate SCells before measurements start. If the **NRCellRelation.InterGnodebFlag** parameter is not set to **TRUE** for any cell, no neighboring cells in gap-assisted measurement reports can be involved in inter-gNodeB CA. This wastes inter-frequency measurements and affects UE throughput in the PCell.

5.1.1.2 Candidate SCell Requirement on Delay

The one-way delay over the eXn interface between the PCell and each SCell for inter-gNodeB CA must meet certain requirements. During candidate SCell selection, the gNodeB evaluates the one-way delay between the PCell and each candidate against the maximum allowed inter-gNodeB one-way delay specified by the **NRDUCellCarrMgmt.InterGnodebDelay** parameter. A cell can be selected as a candidate SCell only when the real-time delay relevant to this cell is less than the configured maximum allowed one-way delay.

For details about how to set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter, see

- [Setting of the Maximum Allowed Inter-gNodeB One-Way Delay for Low-Frequency CA](#) in low-frequency scenarios.
- [Setting of the Maximum Allowed Inter-gNodeB One-Way Delay for Inter-FR CA](#) in inter-FR CA scenarios.

Delay-adaptive downlink inter-gNodeB CA optimizes delay evaluation to improve CA user experience. For details about this function, see [Delay-Adaptive Downlink Inter-gNodeB CA](#).

Setting of the Maximum Allowed Inter-gNodeB One-Way Delay for Low-Frequency CA

Inter-gNodeB CA may or may not take effect, depending on the actual inter-gNodeB delay. The following are the suggestions on how to set the maximum allowed inter-gNodeB one-way delay in different cases of actual delay:

- If the inter-gNodeB delay is less than or equal to 0.5 ms, inter-gNodeB CA works. In this case, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_0DOT5MS**.
- If the inter-gNodeB delay is greater than 0.5 ms and less than or equal to 2 ms, inter-gNodeB CA works. In this case, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_2MS**.
- If the inter-gNodeB delay is greater than 2 ms and less than or equal to 4 ms, inter-gNodeB CA works. In this case, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_4MS**.
- If the inter-gNodeB delay is greater than 4 ms and less than or equal to 8 ms, inter-gNodeB CA works only when fast inter-gNodeB CA (controlled by the **INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter; described in [Downlink Fast Inter-gNodeB CA](#)) is enabled. In this case, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_8MS**.
- If the inter-gNodeB delay is greater than 8 ms, inter-gNodeB CA does not work.

During candidate SCell selection, the gNodeB checks whether the **NRDUCellCarrMgmt.InterGnodebDelay** parameter is set to the same value for the PCell and a candidate. Only cells with the same value as the PCell's value can be selected as candidate SCells.

Inter-gNodeB transmission delay = max(One-way delay, Two-way delay minus one-way delay)

where:

- One-way delay refers to the one-way delay on the user plane of the gNodeB DU eXn interface. It is indicated by the **VS.IPPM.Owd.Means** counter.
- Two-way delay refers to the two-way delay on the user plane of the gNodeB DU eXn interface. It is indicated by the **VS.IPPM.HighLevel.Rtt.Means** counter.

NOTE

The measurement of these counters takes effect on the basis of an IP performance monitoring (IP PM) session whose direction is uplink and differentiated services code point (DSCP) is 0. The **LST IPPMSESSION** command is used to check whether there is an IP PM session that meets the requirements. If there is not, the **ADD IPPMSESSION** command must be executed to add an IP PM session.

Setting of the Maximum Allowed Inter-gNodeB One-Way Delay for Inter-FR CA

For inter-FR CA (enabled by selecting the **INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter), the maximum allowed one-way delay between the low-frequency PCell and inter-gNodeB high-frequency SCells varies as described in [Table 5-1](#), depending on the following functions:

- Function 1: PUCCH grouping adaptation for inter-FR CA, which is controlled by the **PUCCH_GROUP_ADAPT_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter
- Function 2: PUCCH grouping adaptation enhancement for inter-FR CA, which is controlled by the **PUCCH_GROUP_ADAPT_ENH_SW** option of the **NRDUCellCarrMgmt.InterFrCaEnhSw** parameter

Table 5-1 Inter-FR scenarios where inter-gNodeB CA works

Function 1	Function 2	Delay ≤ 0.5 ms	0.5 ms < Delay ≤ 2 ms	2 ms < Delay ≤ 4 ms	Delay > 4 ms
Disabled	/	√	√	√	x
Enabled	Disabled	√	x	x	x
Enabled	Enabled	√	√	√	x
Note: The symbol "√" indicates that inter-gNodeB CA works in the specified scenario, and the symbol "x" indicates the opposite. For details about functions 1 and 2, see <i>Carrier Aggregation</i> .					

In the scenarios where inter-gNodeB CA works, it is recommended that the **NRDUCellCarrMgmt.InterGnodebDelay** parameter be set as follows:

- If the inter-gNodeB transmission delay is less than or equal to 0.5 ms, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_0DOT5MS**.
- If the inter-gNodeB transmission delay is greater than 0.5 ms and less than or equal to 2 ms, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_2MS**.
- If the inter-gNodeB transmission delay is greater than 2 ms and less than or equal to 4 ms, set the **NRDUCellCarrMgmt.InterGnodebDelay** parameter to **DELAY_4MS**.

The inter-gNodeB delay level for inter-gNodeB CA to take effect for a high-frequency SCell can be adaptively selected from 0.5 ms, 2 ms, and 4 ms. It is the same as the level of the actual inter-gNodeB delay for that specific SCell. In this situation, the **NRDUCellCarrMgmt.InterGnodebDelay** parameter can be set to different values for the high-frequency SCells involved in CA. However, a low-frequency PCell and high-frequency SCells cannot be aggregated for inter-gNodeB CA if the actual inter-gNodeB delay between the low-frequency PCell and the high-frequency PU-SCell is greater than the smaller one between the following parameters:

- **NRDUCellCarrMgmt.InterGnodebDelay** for the low-frequency PCell
- **NRDUCellCarrMgmt.InterGnodebDelay** for the high-frequency PU-SCell

Delay-Adaptive Downlink Inter-gNodeB CA

In low-frequency scenarios, Huawei provides delay-adaptive downlink inter-gNodeB CA to further improve CA user experience. When fast inter-gNodeB CA (controlled by the **INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter; described in [Downlink Fast Inter-gNodeB CA](#)) is enabled, delay-adaptive downlink inter-gNodeB CA can be enabled by selecting the **CA_INTER_GNB_DELAY_ADAPT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- With delay-adaptive downlink inter-gNodeB CA enabled: After inter-gNodeB SCells are configured for a UE, the inter-gNodeB delay level for CA to take effect for an SCell is the same as the level of the actual inter-gNodeB delay for that specific SCell. In this situation, the **NRDUCellCarrMgmt.InterGnodebDelay** parameter can be set to different values for the inter-gNodeB cells involved in CA.
- With delay-adaptive downlink inter-gNodeB CA disabled: After inter-gNodeB SCells are configured for a UE, the inter-gNodeB delay level for CA to take effect for all the SCells is the same as the level specified by the **NRDUCellCarrMgmt.InterGnodebDelay** parameter. In this situation, this parameter must be set to an identical value for all the inter-gNodeB cells involved in CA.

Delay-adaptive downlink inter-gNodeB CA involves the inter-gNodeB delay levels in the set of {0.5 ms, 4 ms, 8 ms}. This set can be enriched to further improve CA user experience. In this regard, enhanced delay-adaptive downlink inter-gNodeB CA is introduced. This function improves the flexibility of inter-gNodeB delay level selection. It is controlled by the **CA_INTERGNB_DELAY_ADAPT_2MS_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter.

- With enhanced delay-adaptive downlink inter-gNodeB CA enabled: After inter-gNodeB SCells are configured for a UE, the inter-gNodeB delay level for CA to take effect for an SCell can be 0.5 ms, 2 ms, 4 ms, or 8 ms.
- With enhanced delay-adaptive downlink inter-gNodeB CA disabled: After inter-gNodeB SCells are configured for a UE, the inter-gNodeB delay level for CA to take effect for an SCell can be 0.5 ms, 4 ms, or 8 ms.

NOTE

If the actual inter-gNodeB delay is greater than the value of the **NRDUCellCarrMgmt.InterGnodebDelay** parameter configured on the PCell side, inter-gNodeB CA does not take effect between the PCell and the inter-gNodeB cell, regardless of whether delay-adaptive downlink inter-gNodeB CA is enabled.

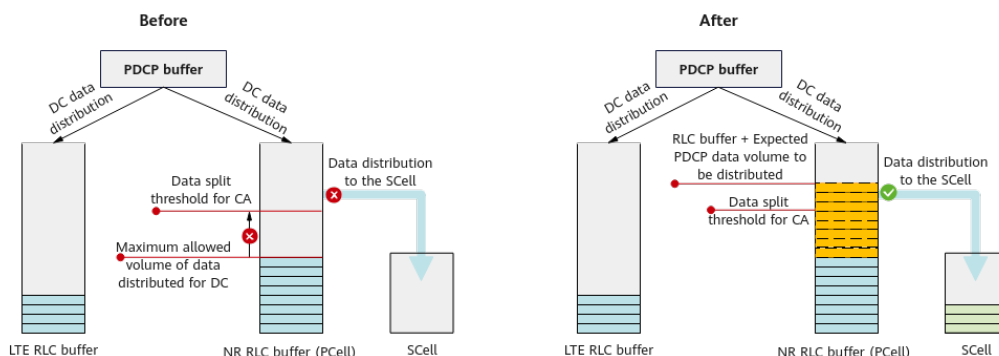
Delay-adaptive downlink inter-gNodeB CA works in inter-FR CA scenarios by default. For details, see [Setting of the Maximum Allowed Inter-gNodeB One-Way Delay for Inter-FR CA](#).

5.1.2 Inter-gNodeB Data Distribution for CA

After activating downlink inter-gNodeB SCells for a CA UE, the serving gNodeB of the PCell distributes downlink data of the UE from the PCell to the SCells. This type of data distribution is triggered only when the data volume in the RLC buffer of the PCell reaches the downlink data split threshold for CA, which is calculated by the serving gNodeB of the PCell. The longer the inter-gNodeB delay, the higher the downlink data split threshold for CA. When the inter-gNodeB delay is long and the NR CA PCell is working in NSA networking, data distribution to SCells can be hardly triggered, affecting CA user experience. That is because in NSA networking, the buffered PDCP data of the NSA DC PCell is distributed to RLC buffers by NSA DC. The amount of data distributed to the NR RLC buffer is subject to an upper limit according to the NSA DC data split rules. In the case of long inter-gNodeB delay, this upper limit is lower than the downlink data split threshold for CA, and therefore data distribution to SCells cannot be triggered.

To address this issue, PDCP-volume-based downlink inter-gNodeB CA is introduced. With this function, the gNodeB considers the data volume in the PDCP buffer expected to be distributed to the NR side, in addition to the data volume in the RLC buffer, when evaluating whether to trigger a data split for inter-gNodeB CA. When the sum of the two values of data volume reaches the downlink data split threshold for CA, data distribution to inter-gNodeB SCells is triggered. This triggering mechanism is more timely and more accurate, improving CA user experience.

Figure 5-1 PDCP-volume-based downlink inter-gNodeB CA



PDCP-volume-based downlink inter-gNodeB CA is controlled by the **PDCP_VOL_BASED_INTER_GNB_CA_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

This function takes effect only when all of the following conditions are met:

- Downlink inter-gNodeB CA is enabled (by selecting the **INTER_GNODEB_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter).
- The PCell is working in NSA Option 3x networking.
- Downlink dynamic data split is used for NSA DC (by setting the **gNBpdcpparamGroup.DlDataPdcpsplitMode** parameter to **SCG_AND_MCG**, **MCG_PREFERRED**, or **ADAPT_PRIMARY_PATH**). For details about downlink dynamic data split, see *EPC-based NSA Basics*.
- The configurations of fast inter-gNodeB CA (controlled by the **INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter) and the maximum inter-gNodeB delay (specified by the **NRDUCellCarrMgmt.InterGnodebDelay** parameter) meet one of the following conditions:
 - Fast inter-gNodeB CA is disabled, and the maximum inter-gNodeB delay is set to be greater than or equal to 4 ms.
 - Fast inter-gNodeB CA is enabled, and the maximum inter-gNodeB delay is set to be greater than or equal to 8 ms.

5.1.3 Functions for Limiting HARQ Resources for CA

Functions for limiting HARQ resources for CA include HARQ resource limitation for CA and adaptive HARQ resource limitation for CA.

- HARQ resource limitation for CA: UEs in the downlink inter-gNodeB CA state take priority over other UEs in requests for PUCCH resources for HARQ feedback. As a result, other UEs will experience a lower success rate of HARQ feedback resource requests. The **CA_HARQ_RES_LIMIT_SW** option of the **NRDUCellCarrMgmt.CaCompatibilitySwitch** parameter can be selected to limit the number of HARQ resources that can be used by UEs in the downlink inter-gNodeB CA state. This reduces the impact of these UEs on the HARQ resource request success rate of other UEs.
- Adaptive HARQ resource limitation for CA: After HARQ resource limitation for CA is enabled, the maximum number of usable HARQ resources becomes fixed for UEs in the downlink inter-gNodeB CA state. This limit does not adapt to changing network conditions. For example, when there are a large number of UEs in the downlink inter-gNodeB CA state, these UEs cannot use resources beyond this fixed limit, leading to resource wastes and affecting their data rates. As such, adaptive HARQ resource limitation for CA is introduced in low-frequency scenarios. This function is controlled by the **CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW** option of the **NRDUCellPucch.PucchAlgoExtSwitch** parameter. After adaptive HARQ resource limitation for CA takes effect, the gNodeB monitors the HARQ resource requests from UEs in the downlink inter-gNodeB CA state over a fixed period of time. It then adaptively increases or decreases the upper limit of HARQ resources available for these UEs as needed. This dynamic

adjustment fully utilizes HARQ resources and guarantees the benefits that can be delivered to the data rates of these UEs.

5.2 Network Analysis

5.2.1 Benefits

- Inter-gNodeB CA enables carrier aggregation across different gNodeBs, expanding the application scenarios of CA. To deliver higher downlink data rates for CA UEs, this function is recommended when the inter-band cell ratio is not 1:1 and the coverage of inter-gNodeB inter-band cells overlaps.
Table 5-2 lists the ranges of peak data rates that a CA UE can achieve using downlink inter-gNodeB 2CC aggregation.

Table 5-2 Ranges of peak data rates achieved using downlink inter-gNodeB 2CC aggregation

SCell Weight Type	Peak Data Rate Range
SRS-based	$(A \times 98\% + B \times 98\%)$ to $(A \times 100\% + B \times 100\%)$
PMI-based	$(A \times 98\% + B \times 95\%)$ to $(A \times 100\% + B \times 100\%)$
Note 1: The SCell weight type depends on CA UE capabilities. If a CA UE transmits SRSs in an SCell, SRS-based weights are used. Otherwise, PMI-based weights are used. Note 2: <i>A</i> is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. Note 3: <i>B</i> is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights.	

 NOTE

If the PCell of a UE uses the dual-period 8:2 slot configuration and an SCell uses the single-period 8:2 slot configuration or the other way round, one slot in the SCell is unavailable for scheduling due to the restriction that no more than eight k1 values can be configured for a single UE in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331. This results in a 1/8 loss in the downlink peak rate of the UE in this SCell, without affecting the downlink peak rate in the PCell. It also affects the overall downlink peak rate of the UE. The amount of the impact varies depending on the PCell and SCell bandwidths. For example, if the two cells have the same bandwidth, there will be a 1/16 loss. The loss in the overall downlink peak rate of the UE is even greater when there are UE compatibility issues.

The compatibility with single-period 8:2 and dual-period 8:2 slot configuration combinations is controlled by the **CA_SLOT_CONFIG_EXT_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter. For details about the slot configurations, see the "Slot Configuration Combinations" section in *Carrier Aggregation*.

- With delay-adaptive downlink inter-gNodeB CA enabled, if the actual inter-gNodeB delay is less than the value of

NRDUCellCarrMgmt.InterGnodebDelay configured on the PCell side, inter-gNodeB CA takes effect based on the actual inter-gNodeB delay. It increases the average downlink throughput of inter-gNodeB CA SCell-activated UEs, which is equal to $(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot}) / N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}$.

- With enhanced delay-adaptive downlink inter-gNodeB CA enabled, if the inter-gNodeB transmission delay is greater than 0.5 ms but less than or equal to 2 ms, inter-gNodeB CA takes effect with the inter-gNodeB delay level of 2 ms. It increases the average downlink throughput of inter-gNodeB CA SCell-activated UEs, which is equal to $(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot}) / N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}$.
- PDCP-volume-based downlink inter-gNodeB CA increases the average downlink throughput of inter-gNodeB CA SCell-activated UEs, which is equal to $(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot}) / N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}$.
- Adaptive HARQ resource limitation for CA increases the average downlink UE throughput (**User Downlink Average Throughput (DU)**) and the average downlink throughput of SCell-activated inter-gNodeB CA UEs, which is equal to $(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot}) / N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}$.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

- When there are uplink services in the PCell, SCell configuration, removal, activation, and deactivation may take effect at slightly different times on the gNodeB and UE sides due to the limited capabilities of some UEs. In this case, the uplink block error rate (BLER) in the PCell slightly increases.
- Initial scheduling begins after an SCell is activated. The duration of this scheduling affects user-perceived data rates in the SCell and is related to factors such as target weights, rank adjustment period, and PCell air interface quality. As a result, the average user-perceived data rates of UEs in SCells decrease in non-full buffer scenarios with frequent SCell deactivation and activation.
- Downlink inter-gNodeB CA decreases the **Cell Downlink Average Throughput (DU)** of SCells and the **User Downlink Average Throughput (DU)** of inter-gNodeB CA UEs in heavy-load scenarios ($70\% \leq \text{cell PRB usage} < 95\%$).
- After downlink inter-gNodeB CA is enabled, PMI-based weights are used for some UEs that treat the local cell as their SCell in heavy-load scenarios. In this case, spectral efficiency is lower than that when SRS-based weights are used. As a result, the **Cell Downlink Average Throughput (DU)** decreases.
- After downlink inter-gNodeB CA is enabled, the CPU usage of the BBP where the PCell is configured increases with the amount of data forwarded from the PCell to SCells at the RLC layer. If the CPU usage is too high, the peak data rate of the UE decreases.

- After delay-adaptive downlink inter-gNodeB CA is enabled, the amount of data distributed to inter-gNodeB cells may decrease if transmission congestion occurs.
- PDCP-volume-based downlink inter-gNodeB CA increases the probability of data distribution from the PCell to SCells. As the NR RLC throughput increases, more data is distributed from the PDCP layer to the NR side, as indicated by the result of $(N.PDCP.Vol.DL.TrfPDU.Tx - N.PDCP.Vol.DL.X2U.TrfPDU.Tx)$. In addition, the amount of data distributed by the PDCP layer to the LTE side remains unchanged or decreases, as indicated by the $N.PDCP.Vol.DL.X2U.TrfPDU.Tx$ counter. As a result, the downlink traffic split rate of LTE ($= N.PDCP.Vol.DL.X2U.TrfPDU.Tx / N.PDCP.Vol.DL.TrfPDU.Tx \times 100\%$) remains unchanged or decreases.
- Adaptive HARQ resource limitation for CA may reduce the number of HARQ resources available to non-inter-gNodeB CA UEs (including intra-gNodeB CA UEs and non-CA UEs), which could lead to decreased throughput for these UEs in extremely heavy-load scenarios.
- In the case of inter-FR CA:
 - Inter-gNodeB inter-FR CA may cause the random access success rate to decrease in the high-frequency cells, if the inter-gNodeB delay is greater than 0.5 ms.
 - After inter-FR CA is enabled, more cell and UE configuration information is exchanged over the eXn interface between base stations. Therefore, the values of counters related to the received and transmitted traffic volumes in the **SCTP Link Measurement** function subset increase.

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051202	Inter-gNodeB CA	NR0S00IPCA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-gNodeB intra-band CA takes effect only when both inter-gNodeB CA (controlled by the INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter) and downlink intra-band CA are enabled.
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-gNodeB intra-FR inter-band CA takes effect only when both inter-gNodeB CA (controlled by the INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter) and downlink intra-FR inter-band CA are enabled.
Low-frequency TDD High-frequency TDD	Inter-FR CA	INTER_FR_FR1_TOFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Inter-gNodeB inter-FR CA takes effect only when both inter-gNodeB CA (controlled by the INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter) and inter-FR CA are enabled.
Low-frequency TDD	Fast inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	- Inter-gNodeB CA with the NRDUCellCarrMgmt.InterGnodebDelay parameter set to DELAY_8MS (with a maximum allowed transmission delay of 8 ms between cells to be aggregated for inter-gNodeB CA) requires fast inter-gNodeB CA to be enabled. - Delay-adaptive downlink inter-gNodeB CA (controlled by the CA_INTER_GNB_DELAY_ADAPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) requires fast inter-gNodeB CA to be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Delay-adaptive downlink inter-gNodeB CA	CA_INTER_GNB_DELAY_ADAPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	Delay-adaptive downlink inter-gNodeB CA must be enabled before enhanced delay-adaptive downlink inter-gNodeB CA (controlled by the CA_INTERGNB_DELAY_ADAPT_2MS_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) can be enabled.
Low-frequency TDD	Downlink inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	PDCP-volume-based downlink inter-gNodeB CA can take effect only if downlink inter-gNodeB CA is enabled.
Low-frequency TDD	Fast inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	PDCP-volume-based downlink inter-gNodeB CA takes effect only when one of the following conditions is met: - Fast inter-gNodeB CA is disabled, and the maximum inter-gNodeB delay is set to be greater than or equal to 4 ms. - Fast inter-gNodeB CA is enabled, and the maximum inter-gNodeB delay is set to be greater than or equal to 8 ms.
Low-frequency TDD	Downlink inter-gNodeB CA	INTER_GNB_ULTRALOW_LAT_SPLT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	PDCP-volume-based downlink inter-gNodeB CA takes effect only when one of the following conditions is met: - Fast inter-gNodeB CA is disabled, and the maximum inter-gNodeB delay is set to be greater than or equal to 4 ms. - Fast inter-gNodeB CA is enabled, and the maximum inter-gNodeB delay is set to be greater than or equal to 8 ms.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUCCH RB adaptation and number of format-3 RBs	- PUCCH RB adaptation (controlled by the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter) - Number of format-3 RBs: NRDUCellPucch.Format3RbNum	<i>Channel Management</i>	Inter-gNodeB CA must be disabled when PUCCH RB adaptation is disabled (that is, the PUCCH_RBRES_ADAPTIVE_SWITCH option of the NRDUCellPucch.PucchAlgoSwitch parameter is deselected) and the NRDUCellPucch.Format3RbNum parameter is set to a value less than RB4.
Low-frequency TDD	HARQ resource limitation for CA	CA_HARQ_RES_LIMIT_SW option of the NRDUCellCarrMgmt.CaCompatibilitySwitch parameter	<i>CA Experience Improvement</i>	Adaptive HARQ resource limitation for CA can take effect only when HARQ resource limitation for CA is enabled.
Low-frequency TDD	Downlink dynamic data split	NsaDcQciParamGroup.DlDataPdcpSplitMode	<i>EPC-based NSA Basics</i>	PDCCP-volume-based downlink inter-gNodeB CA can take effect only if the NsaDcQciParamGroup.DlDataPdcpSplitMode parameter is set to SCG_AND_MCG , MCG_PREFERRED , or ADAPT_PRIMARY_PATH .

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	TRS rate matching optimization	TRS_RATEMATCH_OPT_SW option of the NRDUCellPdsch.RateMatchingSwitch parameter	<i>Downlink Scheduling</i>	Downlink inter-gNodeB CA does not work with TRS rate matching optimization. They cannot be both enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode and NRDUCellMultiTRP.DataTransMode	<i>Cell Combination</i>	Inter-gNodeB CA cannot be enabled in a cell where Joint Transmission Cell Combination has been enabled (by setting NRDUCell.NrDuCellNetworkingMode to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode to JOINT_MODE).
High-frequency TDD	None	None	None	None

NOTE

Apart from what are listed above: Downlink inter-gNodeB intra-band CA shares the mutually exclusive functions of downlink intra-band CA as described in *Carrier Aggregation*. Downlink inter-gNodeB intra-FR inter-band CA shares the mutually exclusive functions of downlink intra-FR inter-band CA as described in *Carrier Aggregation*. Inter-gNodeB inter-FR CA shares the mutually exclusive functions of inter-FR CA as described in *Carrier Aggregation*.

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Smart PDCCH symbol allocation	PDCCH_SYM_SMART_ALLOC_SW option of the NRDUCellPdcch.PdcchAllocationSwitch parameter	<i>Massive MIMO Uplink Boosting (Low-Frequency TDD)</i>	When inter-gNodeB CA takes effect with smart PDCCH symbol allocation enabled for SCells but not the PCell, the average downlink throughput of the inter-gNodeB CA UE may decrease.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Paging code rate adaptation	PAGING_CR_ADAP_SW option of the NRDUCellPagingConfig.PagingSchemeSwitch parameter	<i>5G Networking and Signaling</i>	When inter-gNodeB CA and paging code rate adaptation are both enabled, the gNodeB uses the target paging code rate as defined by the adaptation function to predict the amount of frequency-domain PDSCH resources occupied by paging messages. The predicted amount is greater than that when the adaptation function does not take effect. As a result, the cell has fewer available PDSCH resources remaining.
Low-frequency TDD	Multi-frequency convergent scheduling	MULTI_FREQ_CONVERGENT_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	The more the UEs in the inter-gNodeB CA state, the smaller the gains delivered by multi-frequency convergent scheduling.
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	The more the UEs in the inter-gNodeB CA state, the smaller the gains delivered by CA Performance Boost Based on Multi-dimensional Optimization.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Downlink inter-gNodeB CA does not take effect for 3CC CA UEs for which single DCI has taken effect.
Low-frequency TDD	WUS	DRX_ADAPTATION_SW option of the NRDUCellUEPwrSaving.NrDuCellDrxAlgorithmSwitch parameter	<i>UE Power Saving</i>	WUS does not take effect on a UE with inter-gNodeB CA in effect. If WUS has taken effect on a UE before inter-gNodeB CA, the UE exits WUS before inter-gNodeB CA can take effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Downlink scheduling priority weight factor	gNBDUMacParamGroup.DISchPriWeightFactor or gNBArp.DISchPriWeightFactor	<i>QoS Management</i>	Consider a downlink inter-gNodeB CA scenario where the version of the gNodeB serving the PCell is V100R022C10SPC050 or later and the version of the gNodeB serving the SCell is V100R022C10SPC010 or earlier. If the gNBDUMacParamGroup.DISchPriWeightFactor or gNBArp.DISchPriWeightFactor parameter (specifying the downlink scheduling priority weight factor) is greater than 1000 , the value of $\gamma(QCI) \times Coeff_{RFSP}$ (product of the GUI values) may be greater than 65535000 . In this case, due to the version restriction of the gNodeB serving the SCell, the SCell's effective downlink scheduling priority weight factor (which should use the PCell configuration in a downlink inter-gNodeB CA scenario) is actually less than the PCell's configured value. When it comes to concurrent scheduling of multiple services, the SCell delivers lower data rates to CA UEs

RAT	Function Name	Function Switch	Reference	Description
				than expected based on the PCell's configured scheduling priority weight factor. For details about $\gamma(QCI) \times \text{Coeff}_{RFSP}$, see the section "QoS Guarantee for Downlink Scheduling" in <i>QoS Management</i> .

NOTE

Apart from what are listed above: Downlink inter-gNodeB intra-band CA shares the function impacts of downlink intra-band CA as described in *Carrier Aggregation*. Downlink inter-gNodeB intra-FR inter-band CA shares the function impacts of downlink intra-FR inter-band CA as described in *Carrier Aggregation*. Inter-gNodeB inter-FR CA shares the function impacts of inter-FR CA as described in *Carrier Aggregation*.

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Networking

For downlink inter-gNodeB intra-band CA, the networking requirements described in *Carrier Aggregation* must be fulfilled.

For downlink inter-gNodeB intra-FR inter-band CA, the networking requirements described in *Carrier Aggregation* must be fulfilled.

In addition, downlink inter-gNodeB CA has the following requirements:

- eXn interfaces must be used for inter-gNodeB CA.
 - The **COORD_OUTPUT_BACKHAUL_PT** option of the **BBP.EXFUNCSW** parameter must be selected for all cells to be engaged in inter-gNodeB CA.
 - Xn and eXn interfaces must be set up between the gNodeBs and be working properly. An Xn interface is working properly if **gNodeB CU Xn Interface State** is **Normal** in the output of the **DSP GNBCUXNINTERFACE** command. An eXn interface is working properly if **gNodeB DU eXn User Plane Link State** is **Normal** in the output of the **DSP GNBDUEXNUPINFO** command. For details about how to configure an Xn interface, see *NG and Xn Self-Management*. For details about how to configure an eXn interface, see *eXn Self-Management*.

NOTE

- During initial inter-gNodeB SCell configuration at the local gNodeB, the gNodeB automatically sets up an eXn interface with the peer gNodeB if no eXn interface has been detected. This initial inter-gNodeB SCell configuration will fail. Once the eXn interface is set up, inter-gNodeB SCells can be properly configured.
- In inter-gNodeB inter-FR CA, Xn link self-setup is not supported, and Xn links must be manually set up.
- For details about the requirement on the packet loss rate during transmission, see *Transmission Resource Management*.
- The inter-gNodeB transmission delay must be less than or equal to 4 ms. The inter-gNodeB transmission delay must be less than or equal to 8 ms if fast inter-gNodeB CA (controlled by the **INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter) is enabled.
- Frame offsets must be the same for inter-gNodeB cells in the same frequency band. A frame offset is specified by the **gNBFreqBandConfig.FrameOffset** or **gNodeBParam.FrameOffset** parameter. The **gNBFreqBandConfig.FrameOffset** parameter takes precedence. If it is not configured, the **gNodeBParam.FrameOffset** parameter takes effect.
- The gNodeBs must be time-synchronized to within 1.5 μs. To check the clock quality, run the **STR CLKTST** command to start clock quality monitoring and then run the **DSP CLKTST** command to view the monitoring results.

- The eXn link bandwidth must meet certain requirements additionally posed by inter-gNodeB CA.

The required link bandwidth can be planned based on actual network conditions. The planning results must comply with the transmission bandwidth capacities of the gNodeBs.

For cells with the same bandwidth, the required link bandwidth is calculated as follows:

Link bandwidth = Bandwidth of each SCell x Spectral efficiency of each SCell x Percentage of CA UEs in each SCell x Number of cells to be associated for CA with the gNodeB/Transmission bandwidth coordination efficiency of each SCell

where

- Bandwidth of each SCell: specified by the **NRDUCell.DLBandwidth** parameter. It must be identical for the cells involved in this calculation.
- Spectral efficiency of each SCell: equal to 12 in commercial use scenarios and to 60 in peak cell rate scenarios
- Percentage of CA UEs in each SCell: equal to the result of **(N.User.CA.SCell.DL.Avg/N.User.RRCCConn.Avg x 100%)** in the case of downlink CA. If the counter values are unavailable, the percentage is equal to 60% by default.
- Number of cells to be associated for CA with the gNodeB: number of inter-gNodeB cells to be associated for CA UEs served by the local gNodeB. On commercial networks, it is recommended that this variable be replaced with the number of cells that are served by the local gNodeB and have the same bandwidth as the inter-gNodeB cells. This number is 1 in peak cell rate scenarios, where at most one of the cells served by the gNodeB can reach the peak rate.
- Transmission bandwidth coordination efficiency of each SCell: A bandwidth margin must be reserved considering transport protocol overheads. The default value is 0.8.

The evaluation methods for 2CC aggregation, 3CC aggregation, and massive CA are the same. Take 100 MHz and 60 MHz SCells as an example.

- Assume that it is a commercial use scenario (spectral efficiency is 12), the percentage of CA UEs is 60%, the local gNodeB serves three 100 MHz cells and one 60 MHz cell, and the transmission bandwidth coordination efficiency is 0.8. In this situation, the link bandwidth is equal to 3.24 Gbit/s (= $100 \times 12 \times 60\% \times 3/0.8 + 60 \times 12 \times 60\% \times 1/0.8$).
- Assume that it is a peak cell rate scenario, where at most one of the cells served by the gNodeB can reach the peak rate. For that cell, assume that the bandwidth is 100 MHz, the spectral efficiency is 60, the percentage of CA UEs is 60%, and the transmission bandwidth coordination efficiency is 0.8. In this situation, the link bandwidth is equal to 4.5 Gbit/s (= $100 \times 60 \times 60\% \times 1/0.8$).

Planning the link bandwidth must consider the bandwidths from end to end. If the bandwidth of an intermediate node on the transmission route is insufficient, congestion will occur, increasing the transmission delay and affecting the CA data split efficiency. If the transmission delay is excessive, inter-gNodeB CA cannot take effect.

5.3.5 Others

- For this function to deliver a theoretical peak data rate to SA CA UEs, the maximum bit rate that each CA UE subscribes to in the 5GC cannot be lower than this theoretical value.
- For this NR CA function to deliver a theoretical peak data rate to NSA UEs, the maximum bit rate that each CA UE subscribes to in the EPC cannot be lower than this theoretical value.
- UEs must support CA. During initial access, an incoming handover, or an incoming RRC reestablishment, a UE reports its CA capabilities to the gNodeB.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Before activating this function, ensure that the **NRCellRelation**, **NRCellFreqRelation**, and **NRExternalNCell** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

In addition, ensure that the eXn interface between base stations has been configured. For details about how to configure the eXn interface, see *eXn Self-Management*.

Before enabling downlink inter-gNodeB intra-band CA, enable downlink intra-band CA by preparing data as described in *Carrier Aggregation*.

Before enabling downlink inter-gNodeB intra-FR inter-band CA, enable downlink intra-FR inter-band CA by preparing data as described in *Carrier Aggregation*.

NOTE

If inter-gNodeB CA is enabled before neighboring cells are configured, it does not immediately take effect (up to about 60 minutes later in case of a large number of gNodeBs). This is because data synchronization over the eXn interface takes about 5 minutes between every two gNodeBs.

Table 5-3 describes the parameters used for function activation. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Table 5-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Algorithm Switch	NRDUCellAlgoSwitch.CaAlgorithmSwitch	INTER_GNODEB_CA_SW	Select this option for cells that will participate in inter-gNodeB CA. Before selecting this option, set the NRDUCellCarrMgmt.InterGnodebDelay parameter based on the inter-gNodeB transmission delay and set the NRCCellRelation.InterGnodebFlag parameter to define inter-gNodeB neighboring cells for CA.
Maximum Inter-gNodeB Delay	NRDUCellCarrMgmt.InterGnodebDelay	None	Set this parameter as required. If the PCell has different one-way delays with multiple inter-gNodeB SCells, use the longest of them to set the NRDUCellCarrMgmt.InterGnodebDelay parameter on the PCell side. If the CA_INTER_GNB_DELAY_ADAPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter is deselected, the NRDUCellCarrMgmt.InterGnodebDelay parameter must be set to an identical value for the inter-gNodeB cells that will participate in CA.
Inter-gNodeB Flag	NRCCellRelation.InterGnodebFlag	None	Before enabling inter-gNodeB CA, ensure that the NRCCellRelation.InterGnodebFlag parameter is set to TRUE manually or automatically (depending on the setting of NRCCellAnr.InterGnbCellCfgMode) for the inter-gNodeB neighboring cells allowed to participate in inter-gNodeB CA with the local cell.

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch	CA_INTER_GNB_DELAY_ADAPT_SW	It is recommended that this option be selected. If this option is selected and the delay between the PCell and individual inter-gNodeB SCells differs, set the NRDUCellCarrMgmt.InterGnodebDelay parameter on the PCell side based on the maximum inter-gNodeB one-way delay. The NRDUCellCarrMgmt.InterGnodebDelay parameter can be set to different values for the inter-gNodeB cells involved in CA.
CA Enhanced Algorithm Extended Switch	NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch	CA_INTERGNB_DELAY_ADAPT_2MS_SW	It is recommended that this option be selected. With this option selected, the delay level of 2 ms is available for delay-adaptive inter-gNodeB CA.
CA Compatibility Switch	NRDUCellCarrMgmt.CaCompatibilitySwitch	CA_HARQ_REFS_LIMIT_SW	It is recommended that this option be selected.
PUCCH Algorithm Extension Switch	NRDUCellPucch.PucchAlgorithmExtSwitch	CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW	It is recommended that this option be selected when there are more inter-gNodeB CA UEs than intra-gNodeB CA UEs in a cell.
Inter-gNodeB Cell Config Mode	NRCellAnr.InterGnbCellConfigMode	None	The value AUTO is recommended.
CA Enhanced Algorithm Switch	NRDUCellCarrMgmt.CaEnhancedAlgorithmSwitch	PDPCP_VOL_BASIS_INTER_GNB_CA_SW	It is recommended that this option be selected when downlink dynamic data split is used in the NSA Option 3x architecture and the conditions for the controlled function to take effect are met.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency,

and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

In low-frequency scenarios:

```
//Setting the maximum inter-gNodeB delay (4 ms as an example)
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterGnodebDelay=DELAY_4MS;
//Setting the inter-gNodeB CA flag for a neighboring cell
MOD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0, InterGnodebFlag=TRUE;
//Turning on the inter-gNodeB CA switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
//Enabling delay-adaptive downlink inter-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_INTER_GNB_DELAY_ADAPT_SW-1;
//Enabling enhanced delay-adaptive downlink inter-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_INTERGNB_DELAY_ADAPT_2MS_SW-1;
//Turning on the switch for limiting HARQ feedback resources for CA UEs
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=CA_HARQ_RES_LIMIT_SW-1;
//Enabling adaptive HARQ resource limitation for CA when there are more inter-gNodeB CA UEs than intra-
gNodeB CA UEs in a cell
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW-1;
//Setting the inter-gNodeB cell configuration mode
MOD NRCELLANR: NrCellId=0, InterGnbCellCfgMode=AUTO;
//Enabling PDCP-volume-based downlink inter-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=PDCP_VOL_BASED_INTER_GNB_CA_SW-1;
```

In inter-FR CA scenarios:

```
//Setting the maximum allowed one-way delay between inter-gNodeB cells for CA to 4 ms when PUCCH
grouping adaptation for inter-FR CA is disabled
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterGnodebDelay=DELAY_4MS;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=10, InterGnodebDelay=DELAY_4MS;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
//Setting the maximum allowed one-way delay between inter-gNodeB cells for CA to 0.5 ms when PUCCH
grouping adaptation for inter-FR CA is enabled and PUCCH grouping adaptation enhancement for inter-FR
CA is disabled
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterGnodebDelay=DELAY_0DOT5MS;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=10, InterGnodebDelay=DELAY_0DOT5MS;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
//Setting the maximum allowed one-way delay between inter-gNodeB cells for CA to 4 ms when PUCCH
grouping adaptation for inter-FR CA and PUCCH grouping adaptation enhancement for inter-FR CA are
both enabled
MOD NRDUCELLCARRMGMT: NrDuCellId=0, InterGnodebDelay=DELAY_4MS;
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=10, InterGnodebDelay=DELAY_4MS;
MOD NRDUCELLALGOSWITCH: NrDuCellId=10, CaAlgoSwitch=INTER_GNODEB_CA_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling adaptive HARQ resource limitation for CA
MOD NRDUCELLPUCCH: NrDuCellId=0, PucchAlgoExtSwitch=CA_F3_AND_F4_HARQ_ADAPT_ADJ_SW-0;
```

```
//Disabling HARQ resource limitation for CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaCompatibilitySwitch=CA_HARQ_RES_LIMIT_SW-0;
//Disabling delay-adaptive downlink inter-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_INTER_GNB_DELAY_ADAPT_SW-0;
//Disabling enhanced delay-adaptive downlink inter-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_INTERGNB_DELAY_ADAPT_2MS_SW-0;
//Disabling PDCP-volume-based downlink inter-gNodeB CA
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoSwitch=PDCP_VOL_BASED_INTER_GNB_CA_SW-0;
//Turning off the inter-gNodeB CA switch
MOD NRDUCELLALGOSWITCH: NrDuCellId=0, CaAlgoSwitch=INTER_GNODEB_CA_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Provided that CA UEs are available on the network, downlink inter-gNodeB CA has taken effect if any counter in [Table 5-4](#) returns a non-zero value.

Table 5-4 Counters related to downlink inter-gNodeB CA

Counter ID	Counter Name
1911829360	N.User.IntergNB.CA.PCell.DL.SCellAct.Avg
1911829359	N.User.IntergNB.CA.PCell.DL.SCellAct.Max
1911829362	N.User.IntergNB.CA.PCell.DL.Avg
1911853845	N.User.IntergNB.CA.PCell.DL.Max
1911829361	N.User.IntergNB.CA.SCell.DL.Avg
1911829363	N.User.IntergNB.CA.SCell.DL.Act.Avg

Provided that CA UEs are available on the network, PDCP-volume-based downlink inter-gNodeB CA has taken effect if more data is distributed from the PDCP layer to the NR side as indicated by the result of (**N.PDCP.Vol.DL.TrfPDU.Tx** – **N.PDCP.Vol.DL.X2U.TrfPDU.Tx**).

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 5-5](#).

Table 5-5 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ
1911833674	N.CA.IntergNB.SCell.Add.Att
1911833647	N.CA.IntergNB.SCell.Add.Succ
1911833675	N.CA.IntergNB.SCell.Mod.Att
1911833648	N.CA.IntergNB.SCell.Mod.Succ
1911833676	N.CA.IntergNB.SCell.Rmv.Att
1911833649	N.CA.IntergNB.SCell.Rmv.Succ
1911847072	N.CA.IntergNBDelay.SCell.Rmv.Succ

5.4.3 Network Monitoring

Observe the average downlink UE throughput values to check whether the UE data rate increases after this function is enabled.

- Average downlink UE throughput = **User Downlink Average Throughput (DU)**
- Average downlink throughput of inter-gNodeB SCell-activated CA UEs =
$$\frac{(N.IntergNB.CA.SccActiveUser.ThpVol.DL - N.IntergNB.CA.ThpVol.DL.LastSlot)}{N.IntergNB.CA.ThpTime.DL.RmvLastSlot}$$

NOTE

Optimized measurement of downlink CA throughput can be enabled to improve the accuracy of measuring the average throughput of SCell-activated UEs. For details, see the descriptions of this function in *Carrier Aggregation*.

As inter-gNodeB CA requires the basic CA functions of downlink intra-band CA and downlink intra-FR inter-band CA, the network monitoring method for these basic

CA functions can also be used to monitor the relevant counters. For details about the methods, see section "Network Monitoring" under "Downlink Intra-Band CA" and "Downlink Intra-FR Inter-Band CA" in *Carrier Aggregation*.

6 3CC Aggregation

6.1 Principles

Basic Function of 3CC Aggregation

This function aggregates three intra-FR inter-band or intra-band carriers in the downlink to provide higher bandwidth.

- With downlink intra-FR inter-band CA enabled (by selecting the **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter), only two carriers can be aggregated by default. In this case, three intra-FR inter-band carriers (excluding intra-band carriers) can be aggregated if the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter is set to **DL3CC**.
- With downlink intra-band CA enabled (by selecting the **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter), only two carriers can be aggregated by default. In this case, three intra-band carriers can be aggregated if the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter is set to **DL3CC**.
- With both downlink intra-FR inter-band CA and downlink intra-band CA enabled, only two carriers can be aggregated by default. In this case, three intra-FR inter-band carriers (either including or excluding intra-band carriers) can be aggregated if the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter is set to **DL3CC**.

3CC aggregation is supported only in low frequency bands. For the band +bandwidth combinations for this function, see *Carrier Aggregation*.

Frequency-Priority-based Inter-Frequency Handover Dedicated to 3CC and Massive CA UEs

The function of frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs can be used to improve the experience of 3CC UEs and massive CA UEs. With this function, frequency-priority-based inter-frequency handovers can be performed to transfer 3CC UEs and massive CA UEs to carriers with continuous coverage or better performance. For such a handover, A5

measurement configurations include only the frequencies configured as SCCs before the handover. This prevents gap-assisted measurements during the handover, and therefore no performance loss occurs and the user-perceived rate of the CA UE rises. This function takes effect when both of the following conditions are met:

- Frequency-priority-based inter-frequency handover takes effect. For details about this function, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*. (In *NSA Mobility Management*, this function is known as frequency-priority-based inter-frequency PSCell change.)
- The **3CC_FREQ_PRIORITY_BASED_HO** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected.

This function allows frequency-priority-based inter-frequency handovers to take effect only for 3CC UEs and massive CA UEs.

The procedure for a frequency-priority-based inter-frequency handover for a 3CC UE or a massive CA UE is the same as that for a frequency-priority-based inter-frequency handover described in *SA Mobility Management in Connected Mode* and that for a frequency-priority-based inter-frequency PSCell change described in *NSA Mobility Management*.

Reselection Policy for 3CC and Massive CA UEs in Idle Mode

If a UE reselects a cell that does not support the 3CC or massive CA function, the function cannot take effect for the UE. To increase the proportion of UEs with 3CC or massive CA in effect, the "reselection policy for 3CC and massive CA UEs in idle mode" function is introduced in SA networking. This function is controlled by the **MASS_CA_IDLE_RESEL_POL_SW** option of the **NRCellOpQciBearer.ServiceDiffAlgoSwitch** parameter. With this function enabled, the gNodeB gives top dedicated reselection priority to the target PCC frequency supporting 3CC and massive CA during cell reselection for capable UEs with specific operator and QCI details. This increases the probability that these UEs can be reselected to cells capable of 3CC and massive CA, thereby increasing the proportion of UEs with 3CC and massive CA in effect.

There are multiple types of dedicated reselection priorities. If they are configured concurrently, they are delivered in different sequences. For details about the sequence in which dedicated reselection priorities are delivered and the basic information about cell reselection, see *SA Mobility Management in Idle Mode*.

Downlink RLC Throughput Distribution Measurement for 3CC or Higher CA UEs

If the downlink RLC throughput distributions of CA UEs with three or more CCs aggregated (referred to as 3CC or higher CA UEs in this document) are to be measured, the **NRDUCellOamParam.UeSpecStatisticsPolicy** parameter can be set to **DL_NCC_UE** to enable the "downlink RLC throughput distribution measurement for 3CC or higher CA UEs" function. After this function is enabled, the counters **N.Thp.DL.Ue.Spec.Samp.Index0** to **N.Thp.DL.Ue.Spec.Samp.Index14** can be used to observe the downlink RLC throughput distributions of 3CC or higher CA UEs. For details about the counters **N.Thp.DL.Ue.Spec.Samp.Index0** to **N.Thp.DL.Ue.Spec.Samp.Index14** and the settings of throughput distribution ranges, see the description of the corresponding counters.

6.2 Network Analysis

6.2.1 Benefits

The downlink peak data rates of CA UEs increase.

Table 6-1 lists the ranges of peak data rates that a CA UE can achieve using downlink 3CC aggregation.

Table 6-1 Ranges of peak data rates achieved using downlink 3CC aggregation

SCell Weight Type	Peak Data Rate Range
SRS-based	$(A \times 98\% + B \times 98\% + C \times 98\%)$ to $(A \times 100\% + B \times 100\% + C \times 100\%)$
PMI-based	$(A \times 98\% + B \times 95\% + C \times 95\%)$ to $(A \times 100\% + B \times 100\% + C \times 100\%)$
Note 1: The SCell weight type depends on CA UE capabilities. If a CA UE transmits SRSs in an SCell, SRS-based weights are used. Otherwise, PMI-based weights are used. Note 2: <i>A</i> is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. Note 3: <i>B</i> is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights. Note 4: <i>C</i> is the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights.	

 NOTE

If the PCell of a UE uses the dual-period 8:2 slot configuration and an SCell uses the single-period 8:2 slot configuration or the other way round, one slot in the SCell is unavailable for scheduling due to the restriction that no more than eight k1 values can be configured for a single UE in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331. This results in a 1/8 loss in the downlink peak rate of the UE in this SCell, without affecting the downlink peak rate in the PCell. It also affects the overall downlink peak rate of the UE. The amount of the impact varies depending on the PCell and SCell bandwidths. For example, if the two cells have the same bandwidth, there will be a 1/16 loss. The loss in the overall downlink peak rate of the UE is even greater when there are UE compatibility issues.

The compatibility with single-period 8:2 and dual-period 8:2 slot configuration combinations is controlled by the **CA_SLOT_CONFIG_EXT_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter. For details about the slot configurations, see the "Slot Configuration Combinations" section in *Carrier Aggregation*.

After 3CC aggregation is enabled, the average throughput of SCell-activated CA UEs in the 3CC aggregation state may not increase in scenarios with a relatively large number of small-packet services. That is because the protocol-defined method of calculating this throughput does not count in tail packets, but 3CC

aggregation widens the bandwidth for CA UEs, boosts their scheduling, and increases the proportion of tail packets in these scenarios. The average throughput of SCell-activated CA UEs in the 3CC aggregation state is equal to $(N.3CC.CA.ThpVol.DL - N.3CC.CA.ThpVol.DL.LastSlot) / N.3CC.CA.ThpTime.DL.RmvLastSlot$.

After the "reselection policy for 3CC and massive CA UEs in idle mode" function is enabled, the cell may have an increased proportion of 3CC and massive CA UEs with specific operator and QCI details, and there may be an increase in the average throughput of SCell-configured CA UEs $((N.CA.SccConfigUser.ThpVol.DL - N.CA.SccConfigUser.ThpVol.DL.LastSlot) / N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot)$.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

The network impacts of 3CC aggregation are the same as those of downlink intra-FR inter-band CA and downlink intra-band CA. For details, see *Carrier Aggregation*.

If uplink enhancement for smart carrier selection takes effect in suitable scenarios, the number of UEs for which 3CC CA takes effect decreases. As a result, the average number of downlink CA UEs that treat the local cell as their PCell ($N.User.CA.PCell.DL.Avg$) decreases. For details about uplink enhancement for smart carrier selection, see *Carrier Aggregation*.

6.3 Requirements

6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-051211	3CC CA	NR0SDL3CCA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Intra-FR inter-band 3CC aggregation (excluding intra-band carriers) requires downlink intra-FR inter-band CA to be enabled.
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAIgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- Intra-band 3CC aggregation requires downlink intra-band CA to be enabled. - If 3CC aggregation involves both intra-band and intra-FR inter-band carriers, the related prerequisite function(s) must be enabled for 3CC aggregation to take effect: (1) the downlink intra-band CA function for a cell participating in intra-band CA, (2) the downlink intra-FR inter-band CA function for a cell participating in intra-FR inter-band CA, or (3) both the downlink intra-band CA and downlink intra-FR inter-band CA functions for a cell participating in both CA types.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs can take effect only when frequency-priority-based inter-frequency handover is in effect.
Low-frequency TDD	Frequency-priority-based inter-frequency PSCell change	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>NSA Mobility Management</i>	Frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs can take effect only when frequency-priority-based inter-frequency PSCell change is in effect.
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Channel Management</i>	The SRS_INTRF_COORD_S_SLOT_SW option of this parameter must be selected if (1) NRDUCellCarrMgmt.CaDLMaxCcNum is set to DL3CC/DL4CC/DL5CC and (2) the switch for SRS remote interference avoidance is turned on.
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	

6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Joint Transmission Cell Combination	NRDUCell.NrDuCellNetworkingMode and NRDUCellMultiTRP.DataTransMode	<i>Cell Combination</i>	3CC aggregation cannot be enabled in a cell where Joint Transmission Cell Combination has been enabled (by setting NRDUCell.NrDuCellNetworkingMode to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTRP.DataTransMode to JOINT_MODE).
Low-frequency TDD	Compensation for SAR decrease (low-frequency TDD)	REDUCE_SAR_COMPENSATION_SW option of the NRDUCellUeCoop.SpecUeCooperationSwitch parameter	<i>Device-Pipe Synergy</i>	CA does not work with compensation for SAR decrease.
Low-frequency TDD	SRS interference coordination based on self-contained and uplink slots	SRS_INTRF_COORD_S_U_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Channel Management</i>	SRS interference coordination based on self-contained and uplink slots cannot be enabled when the NRDUCellCarrMgmt.CaDlMaxCcNum parameter is set to DL3CC/DL4CC/DL5CC . For details about scenarios where NRDUCellCarrMgmt.CaDlMaxCcNum is set to DL4CC/DL5CC , see 8.1 Principles .

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low latency and high reliability	HIGH_RELIABILITY_BASIC_SW option of the NRDUCellIgoSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	CA does not work with the function of low latency and high reliability.

6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	32-port CSI-RS measurement on SCCs	CA_SCC_CSI_RS_32PORT_SW option of the NRDUCellCsiRs.CsiSwitch parameter	<i>Carrier Aggregation</i>	32-port CSI-RS measurement on SCCs does not take effect for CA UEs for which 3CC aggregation is in effect.
Low-frequency TDD	Hyper Cell Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell Cell Combination</i>	The cells involved in 3CC aggregation cannot include both hyper cells and combined cells. (Combined cells are those with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to BASIC_MODE or DPS_MODE).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink 8-layer transmission for a single FWA user	DL_SU_EIGHT_LAYER_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	- When the DL_EIGHT_LAYER_SCENARIO_EXT_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter is deselected, downlink 8-layer transmission for a single FWA user cannot take effect together with 3CC aggregation for a UE. If 3CC aggregation is enabled, enabling downlink 8-layer transmission for a single FWA user may decrease the success rate of SCell configuration for 3CC aggregation. - When the DL_EIGHT_LAYER_SCENARIO_EXT_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter is selected, downlink 8-layer transmission for a single FWA user can take effect for 3CC or higher CA UEs.
Low-frequency TDD	Downlink 8-layer scenario extension	DL_EIGHT_LAYER_SCENARIO_EXT_SW option of the NRDUCellDLRank.DlRankSelAlgoSw parameter	<i>FWA</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Proactive avoidance of remote interference	SRS_RIM_INTRF_AVOID_SW and SRS_INTRF_COORD_S_SLOT_SW options of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Remote Interference Management (Low-Frequency TDD)</i>	After proactive avoidance of remote interference is enabled, more PUCCH RBs will be used in the PCell.
Low-frequency TDD	T320 timer	NRCelReselConfig.T320ForOther	<i>SA Mobility Management in Idle Mode</i>	If both dedicated-priority-based idle mode MLB and the reselection policy for 3CC and massive CA UEs in idle mode are enabled, the T320 timer takes the value of NRCelReselConfig.T320ForOther for 3CC and massive CA UEs.
Low-frequency TDD	Dedicated-priority-based idle mode MLB	All of the following settings apply: NRCelMLbIdleMLbPolicy set to DEDICATED_PRIORITY INTER_FREQ_IDLE_MLB_SW option of the NRCelAlgoSwitch.MlbAlgoSwitch parameter selected	<i>Mobility Load Balancing in Idle Mode</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Frequency-priority-based inter-frequency handover	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	After frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs takes effect, frequency-priority-based inter-frequency handover does not take effect for non-3CC UEs or non-massive-CA UEs.
Low-frequency TDD	Frequency-priority-based inter-frequency PSCell change	FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>NSA Mobility Management</i>	After frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs takes effect, frequency-priority-based inter-frequency PSCell change does not take effect for non-3CC UEs or non-massive-CA UEs.
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRIORITY_BASED_INTER_FREQ_HO_SWITCH option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	When both "frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs" and "CA-UE-specific frequency-priority-based inter-frequency handover" are enabled, only the latter takes effect.

NOTE

3CC aggregation requires downlink intra-band CA and downlink intra-FR inter-band CA. Apart from what are listed above, 3CC aggregation shares the function impacts of downlink intra-band CA and downlink intra-FR inter-band CA as described in *Carrier Aggregation*.

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

For downlink 3CC aggregation:

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

CA can be performed in one baseband processing unit or between baseband processing units.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Networking

Other networking requirements of 3CC aggregation are the same as those of downlink intra-FR inter-band CA and downlink intra-band CA. For details, see *Carrier Aggregation*.

6.3.5 Others

As described in [6.3.2.1 Prerequisite Functions](#), 3CC aggregation requires downlink intra-band CA or downlink intra-FR inter-band CA.

For other requirements of downlink intra-band 3CC aggregation, see the "Others" section for downlink intra-band CA in *Carrier Aggregation*. The other requirements of downlink intra-FR inter-band 3CC aggregation slightly differ from those of downlink intra-FR inter-band CA. Specifically:

- For this function to deliver a theoretical peak data rate to SA CA UEs, the maximum bit rate that each CA UE subscribes to in the 5GC cannot be lower than this theoretical value.

- For this NR CA function to deliver a theoretical peak data rate to NSA UEs, the maximum bit rate that each CA UE subscribes to in the EPC cannot be lower than this theoretical value.
- UE requirements:
 - UEs must support CA. During initial access, an incoming handover, or an incoming RRC reestablishment, a UE reports its CA capabilities to the gNodeB.
 - For CA with both the PCell and SCells being NR TDD cells with different slot configurations or CA with the PCell and at least one SCell being in different duplex modes, UEs must support simultaneous transmission and reception. That is, UEs must report the *SimultaneousRxTxInterBandCA* and *SimultaneousRxTxInterBandCAPerBandPair* fields. Otherwise, 3CC aggregation cannot take effect.
 - UEs must support semi-static codebooks (as indicated by *semiStaticHARQ-ACK-Codebook* in the *phy-Parameters* IE) for feedback over the PUCCH.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

Before activating this function, ensure that the **NRCellRelation**, **NRCellFreqRelation**, and **NRExternalNCell** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

Table 6-2 describes the parameters used for function activation. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Before enabling intra-band 3CC aggregation, enable downlink intra-band CA by preparing data as described in *Carrier Aggregation*.

Before enabling intra-FR inter-band 3CC aggregation, enable downlink intra-FR inter-band CA by preparing data as described in *Carrier Aggregation*.

Table 6-2 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Band 1	gNBCustCaB and Comb.FrequencyBand1	N/A	This parameter specifies CA band 1. Set this parameter as required.

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Bandwidth 1	gNBCustCaB andComb.FrequencyBandwidth1	N/A	This parameter specifies the bandwidth of CA band 1. Set this parameter as required.
Frequency Band 2	gNBCustCaB andComb.FrequencyBand2	N/A	This parameter specifies CA band 2. Set this parameter as required.
Frequency Bandwidth 2	gNBCustCaB andComb.FrequencyBandwidth2	N/A	This parameter specifies the bandwidth of CA band 2. Set this parameter as required.
Frequency Band 3	gNBCustCaB andComb.FrequencyBand3	N/A	This parameter specifies CA band 3. Set this parameter as required.
Frequency Bandwidth 3	gNBCustCaB andComb.FrequencyBandwidth3	N/A	This parameter specifies the bandwidth of CA band 3. Set this parameter as required.
CA Downlink Max CC Number	NRDUCellCarMgmt.CaDlMaxCcNum	N/A	To enable downlink 3CC aggregation, set this parameter to DL3CC .
Inter-Frequency Handover Switch	NRCelAlgoSwitch.InterFreqHoSwitch	3CC_FREQ_PRIORITY_BASED_HO	Select this option if frequency-priority-based inter-frequency handover (in SA networking) or frequency-priority-based inter-frequency PSCell change (in NSA networking) is expected to take effect only for 3CC UEs and massive CA UEs.
Service Differentiated Algorithm Switch	NRCelOpQciBearer.ServiceDiffAlgoSwitch	MASS_CA_IDLE_RESEL_POLICY_SW	It is recommended that this option be selected if the proportion of 3CC and massive CA UEs with specific operator and QCI details is expected to be increased.
UE-specific Statistics Policy	NRDUCellOamParam.UeSpecificStatisticsPolicy	N/A	If the downlink RLC throughput distributions of 3CC or higher CA UEs are to be measured, set NRDUCellOamParam.UeSpecificStatisticsPolicy to DL_NCC_UE .

Parameter Name	Parameter ID	Option	Setting Notes
NR Smart Multi-Carrier Switch	gNodeBAlgo. NrSmartMultiCarrSw	CA_CAP_USER_STATS_POL_SW	Select this option if it is expected that CA-capable UE counters measure only the numbers of UEs that can actually work in the capability-indicated 3CC aggregation state with the present cell configurations.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the maximum number of downlink carriers to enable downlink 3CC aggregation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDiMaxCcNum=DL3CC;
//Optional; TDD) Adding a customized CA band+bandwidth combination
ADD GNBCUSTCABANDCOMB: CustCaBandCombId=1, BandwidthCombSet=0, FrequencyBand1=N77,
FrequencyBandwidth1=CELL_BW_100M, FrequencyBand2=N77, FrequencyBandwidth2=CELL_BW_90M,
FrequencyBand3=N41, FrequencyBandwidth3=CELL_BW_100M;
//Turning on 3CC_FREQ_PRIORITY_BASED_HO if frequency-priority-based inter-frequency handover (in SA
networking) or frequency-priority-based inter-frequency PSCell change (in NSA networking) is expected to
take effect only for 3CC UEs and massive CA UEs
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=3CC_FREQ_PRIORITY_BASED_HO-1;
//Optional) Turning on MASS_CA_IDLE_RESEL_POL_SW if the proportion of 3CC and massive CA UEs with
specific operator and QCI details is expected to be increased
ADD NRCELLOPQCIBEARER: NrCellId=0, OperatorId=0, Qci=0,
ServiceDiffAlgoSwitch=MASS_CA_IDLE_RESEL_POL_SW-1;
//Optional) Setting UeSpecStatisticsPolicy to DL_NCC_UE if the downlink RLC throughput distributions of
3CC or higher CA UEs are to be measured
MOD NRDUCELLLOAMPARAM: NrDuCellId=0, UeSpecStatisticsPolicy=DL_NCC_UE;
//Optional) Enabling optimized measurement of the numbers of CA-capable UEs in a cell if it is expected
that CA-capable UE counters measure only the numbers of UEs that can actually work in the capability-
indicated 3CC aggregation state with the present cell configurations
MOD GNODEBALGO: NrSmartMultiCarrSw=CA_CAP_USER_STATS_POL_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink 3CC aggregation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDiMaxCcNum=NOT_CONFIG;
//Turning off 3CC_FREQ_PRIORITY_BASED_HO
```

```
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=3CC_FREQ_PRIORITY_BASED_HO-0;  
//Turning off MASS_CA_IDLE_RESEL_POL_SW  
MOD NRCELLOPQCIBEARER: NrCellId=0, OperatorId=0, Qci=0,  
ServiceDiffAlgoSwitch=MASS_CA_IDLE_RESEL_POL_SW-0;  
//Disabling optimized measurement of the numbers of CA-capable UEs in a cell  
MOD GNODEBALGO: NrSmartMultiCarrSw=CA_CAP_USER_STATS_POL_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

Provided that CA UEs are available on the network, 3CC aggregation has taken effect if any counter in [Table 6-3](#) returns a non-zero value. These counters are related to the downlink data volume and data transmission duration regarding downlink 3CC aggregation.

Table 6-3 Counters related to downlink 3CC aggregation

Counter ID	Counter Name
1911833653	N.3CC.CA.ThpVol.DL
1911833651	N.3CC.CA.ThpVol.DL.LastSlot
1911833654	N.3CC.CA.ThpTime.DL.RmvLastSlot
1911844995	N.User.CA.PCell.DL.3CC.Avg
1911853843	N.User.CA.PCell.DL.3CC.Max
1911849170	N.User.CA.PCell.DL.3CC.SCellAct.Avg

The counters related to the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal are listed in [Table 6-4](#).

Table 6-4 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att

Counter ID	Counter Name
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

6.4.3 Network Monitoring

Monitor the **N.3CC.CA.ThpVol.DL** counter to observe the RLC traffic volume to UEs in the 3CC aggregation state in a cell. This counter measures the total volume of downlink data sent at the RLC layer in a cell to SCell-activated CA UEs in the 3CC aggregation state.

Observe the average throughput of SCell-activated UEs in the 3CC aggregation state, which is calculated using the following formula: **(N.3CC.CA.ThpVol.DL – N.3CC.CA.ThpVol.DL.LastSlot)/N.3CC.CA.ThpTime.DL.RmvLastSlot**.

 NOTE

Optimized measurement of downlink CA throughput can be enabled to improve the accuracy of measuring the average throughput of SCell-activated UEs. For details, see the descriptions of this function in *Carrier Aggregation*.

As 3CC aggregation requires the basic CA functions of downlink intra-band CA and downlink intra-FR inter-band CA, the network monitoring method for these basic CA functions can also be used to monitor the relevant counters. For details about the methods, see section "Network Monitoring" under "Downlink Intra-Band CA" and "Downlink Intra-FR Inter-Band CA" in *Carrier Aggregation*.

7 Fast CA

7.1 Principles

The following fast CA functions are now available: downlink fast intra-gNodeB CA, downlink fast inter-gNodeB CA, and uplink fast intra-gNodeB CA.

- For details about downlink fast intra-gNodeB CA, see [Downlink Fast Intra-gNodeB CA](#).
- For details about downlink fast inter-gNodeB CA, see [Downlink Fast Inter-gNodeB CA](#).
- For details about uplink fast intra-gNodeB CA, see [Uplink Fast Intra-gNodeB CA](#).

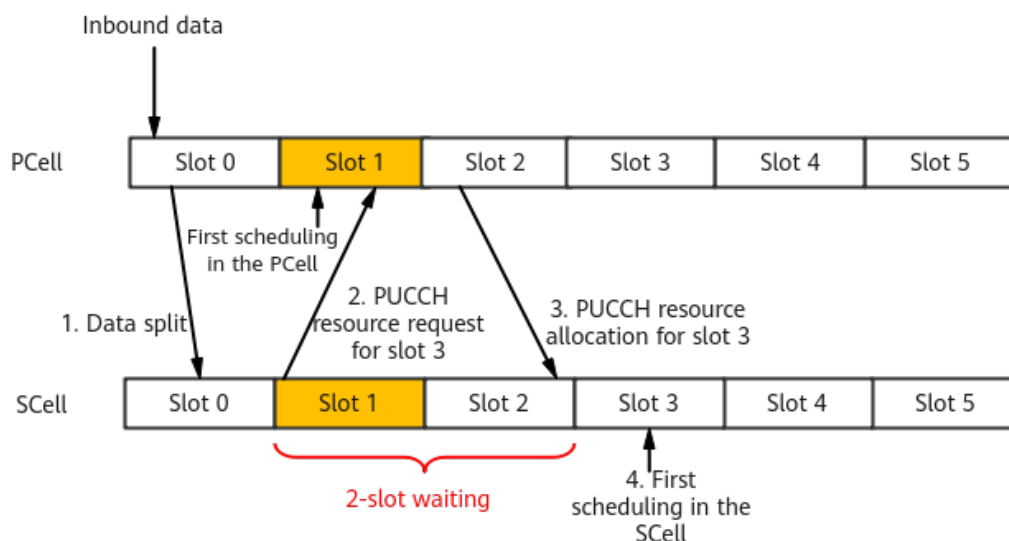
Downlink Fast Intra-gNodeB CA

As shown in [Figure 7-1](#), after an SCell is activated for a CA UE, the following procedure occurs before first downlink scheduling in the SCell:

1. The gNodeB serving the PCell distributes part of the data buffered at the RLC layer to the SCell.
2. The SCell requests PUCCH resources from the PCell.
3. After receiving the request from the SCell, the PCell allocates PUCCH resources to the SCell.
4. Downlink scheduling begins in the SCell.

Consequently, the first downlink scheduling of the UE begins two to three slots later than in the PCell, leading to non-optimal experience of the CA UE.

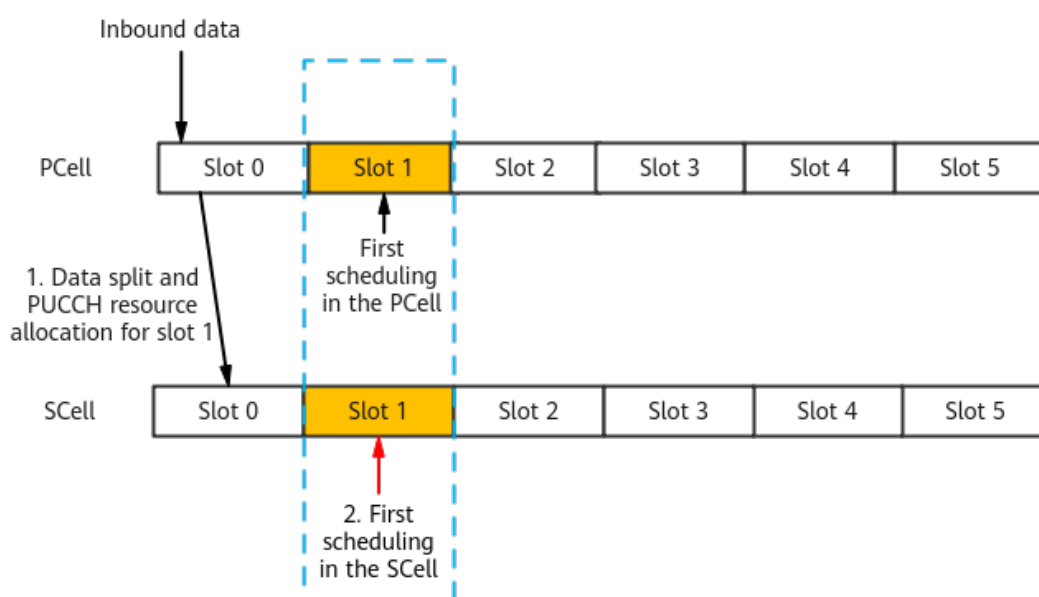
Figure 7-1 Downlink non-fast intra-gNodeB CA



To optimize the experience of intra-gNodeB CA UEs during data splits, downlink fast intra-gNodeB CA is introduced. As illustrated in [Figure 7-2](#), downlink scheduling in an activated SCell of a CA UE can begin as soon as the SCell receives the data distributed from the PCell. The detailed procedure is as follows:

1. When the buffered RLC data volume at the PCell is greater than or equal to the downlink threshold for a CA data split, the gNodeB serving the PCell distributes part of the data buffered at the RLC layer to the SCell and allocates PUCCH resources to the SCell.
2. After the SCell receives the data distributed from the PCell, downlink scheduling can begin in the SCell, so that data transmission starts simultaneously in the SCell and PCell.

Figure 7-2 Downlink fast intra-gNodeB CA



Downlink fast intra-gNodeB CA works only in low frequency bands. For details about supported band+bandwidth combinations, see *Carrier Aggregation*. In addition, downlink fast intra-gNodeB CA works only between intra-band or intra-FR inter-band carriers served by the same gNodeB in the downlink. This function is controlled by the **CA_ULTRA_LOW_LATENCY_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter. It takes effect so long as this option is selected for the PCell.

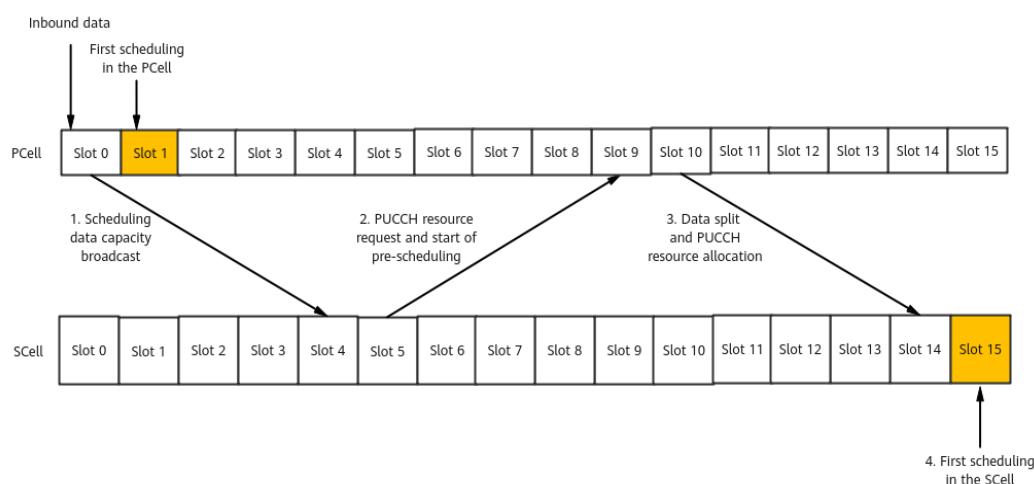
Downlink Fast Inter-gNodeB CA

As shown in [Figure 7-3](#), after an SCell is activated for a CA UE, the following procedure occurs before first downlink scheduling in the SCell:

1. The PCell obtains the scheduling data capacity allocated to each SCell based on the scheduling capabilities of individual SCells and broadcasts the result to each SCell.
2. After an SCell receives the scheduling data capacity broadcast by the PCell, pre-scheduling starts, estimating the amount of data that can be scheduled in the SCell in twice the inter-gNodeB transmission delay. The SCell sends the estimation result to the proxy RLC buffer module associated with itself and then requests PUCCH resources from the PCell. For details about the inter-gNodeB transmission delay, see [5.1.1.2 Candidate SCell Requirement on Delay](#).
3. After receiving the request from the SCell, the gNodeB serving the PCell distributes the data buffered at the RLC layer to the SCell and allocates PUCCH resources that would be appropriate in one time the inter-gNodeB transmission delay.
4. Downlink scheduling in the SCell starts at the specified pre-scheduling time.

Consequently, the first downlink scheduling of the UE begins about three times the inter-gNodeB transmission delay later than in the PCell, leading to non-optimal experience of the CA UE.

Figure 7-3 Downlink non-fast inter-gNodeB CA

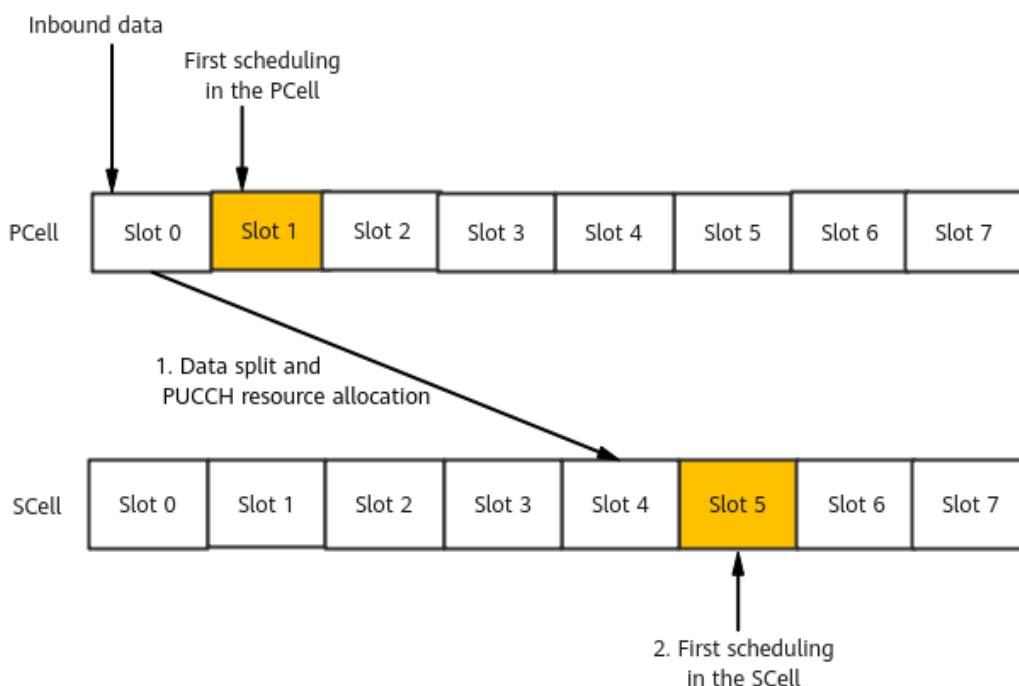


To optimize the experience of inter-gNodeB CA UEs during data splits, downlink fast inter-gNodeB CA is introduced. As illustrated in [Figure 7-4](#), downlink

scheduling in an activated SCell of a CA UE can begin as soon as the SCell receives the data distributed from the PCell. The detailed procedure is as follows:

1. The PCell obtains the data capacity allocated to each SCell based on the scheduling capabilities of individual SCells and the CA data split algorithm. Meanwhile, the PCell distributes the data buffered at the RLC layer to each SCell based on the data capacity and allocates PUCCH resources to each SCell accordingly.
2. After an SCell receives the data distributed from the PCell, downlink scheduling can begin in the SCell.

Figure 7-4 Downlink fast inter-gNodeB CA



Downlink fast inter-gNodeB CA works only in low frequency bands. For details about supported band+bandwidth combinations, see *Carrier Aggregation*. This function can be enabled only when downlink inter-gNodeB CA between intra-band or intra-FR inter-band carriers is enabled. Downlink fast inter-gNodeB CA is controlled by the **INTER_GNB_ULTRALOW_LAT_SPLT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch** parameter. It takes effect only when this option is selected for both the PCell and SCells.

Uplink Fast Intra-gNodeB CA

After an uplink SCell is activated, the first scheduling can be performed only after the gNodeB determines that the channel quality of the SCell meets the transmission requirements. Without uplink fast intra-gNodeB CA, determining the SCell's channel quality is relatively time-consuming, delaying the first scheduling in the SCell. In uplink intra-band CA and uplink intra-FR inter-band CA scenarios, uplink fast intra-gNodeB CA is introduced to optimize the procedure for determining the SCell channel quality and shorten the decision-making time. This

enables faster first scheduling in the SCell, improving CA user experience. This function is controlled by the **UL_CA_FAST_SPLIT_SW** option of the **NRDUCellCarrMgmt.CaEnhancedAlgoSwitch** parameter.

7.2 Network Analysis

7.2.1 Benefits

After downlink fast intra-gNodeB CA is enabled, the average throughput of SCell-activated CA UEs increases by about 10% if the ratio of the SCell bandwidth (specified by **NRDUCell.DlBandwidth**) to the PCell bandwidth is large (for example, greater than or equal to 4). This average throughput equals $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$. The more superior the scheduling capability of an SCell is to that of the PCell, the greater the increase, while the longer the continuous scheduling duration, the smaller the increase. In full buffer scenarios, only small gains or even no gains are yielded.

After downlink fast inter-gNodeB CA is enabled, the average throughput of SCell-activated inter-gNodeB CA UEs increases by 5%–20% if the ratio of the SCell bandwidth (specified by **NRDUCell.DlBandwidth**) to the PCell bandwidth is large (for example, greater than or equal to 4). This average throughput equals $(N.IntergNB.CA.SccActiveUser.ThpVol.DL - N.IntergNB.CA.ThpVol.DL.LastSlot) / N.IntergNB.CA.ThpTime.DL.RmvLastSlot$. In addition, the average throughput of SCell-activated CA UEs may increase. This average throughput equals $(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot) / N.CA.ThpTime.DL.RmvLastSlot$. The more superior the scheduling capability of an SCell is to that of the PCell, the greater the increase. The longer the continuous scheduling duration, the smaller the increase. In full buffer scenarios, only small gains or even no gains are yielded.

Uplink fast intra-gNodeB CA enables faster first scheduling in SCells so that data can be scheduled earlier. This may increase the average throughput of SCell-activated uplink CA UEs. This average throughput equals $(N.CA.SccActiveUser.ThpVol.UL - N.CA.ThpVol.UL.SmallPkt) / N.CA.ThpTime.UL.RmvSmallPkt$. The gains in throughput improvement decrease as the continuous scheduling duration lasts longer.

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

- After downlink fast intra- or inter-gNodeB CA is enabled, the sizes of tail packets in the PCell and SCells may change. If this is the case, the average MCS indexes in the PCell and SCells fluctuate, as the MCS indexes for small packets are lowered.
- After downlink fast intra- or inter-gNodeB CA is enabled, the PCell distributes data to SCells and allocates PUCCH resources for the SCells in advance. As a result, the PCell throughput $(N.ThpVol.DL.Cell / N.ThpTime.DL.Cell)$ decreases, and the average cell throughput of the entire network changes. In addition, as scheduling starts earlier in an SCell, the average number of

scheduled UEs ($\text{N.User.Schedule.DL.Sum} / \text{N.ThpTime.DL.Cell} \times 1000$) changes.

- After downlink fast inter-gNodeB CA is enabled, the PCell distributes data to SCells in advance. The data is not counted in the statistics on the average downlink throughput of other UEs in the SCells, but distributing the data in advance occupies SCell resources. Therefore, the average downlink throughput of other UEs in the SCells decreases. In general, there are no negative impacts on the average value of the average downlink UE throughput in the PCell and SCells. However, this average value alters in heavy-load scenarios, where the PRB usage is greater than or equal to 70%. Average downlink UE throughput = $(\text{N.ThpVol.DL} - \text{N.ThpVol.DL.LastSlot}) / \text{N.ThpTime.DL.RmvLastSlot}$

7.3 Requirements

7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-061224	Fast CA	NR0S0FASTCA0	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-band fast intra-gNodeB CA (controlled by the CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) requires that downlink intra-band CA be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Downlink intra-FR inter-band fast intra-gNodeB CA (controlled by the CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) requires that downlink intra-FR inter-band CA be enabled.
Low-frequency TDD	Inter-gNodeB CA	INTER_GNODEB_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>CA Experience Improvement</i>	Downlink fast inter-gNodeB CA (controlled by the INTER_GNB_ULTRALOW_LAT_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) requires that inter-gNodeB CA be enabled.
Low-frequency TDD	Downlink fast intra-gNodeB CA	CA_ULTRA_LOW_LATENCY_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>CA Experience Improvement</i>	Downlink fast inter-gNodeB CA (controlled by the INTER_GNB_ULTRALOW_LAT_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter) requires that downlink fast intra-gNodeB CA be enabled.
Low-frequency TDD	Uplink intra-band CA	INTRA_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-band fast intra-gNodeB CA (controlled by the UL_CA_FAST_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that uplink intra-band CA be enabled.
Low-frequency TDD	Uplink intra-FR inter-band CA	INTRA_FR_INTER_BAND_UL_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	Uplink intra-FR inter-band fast intra-gNodeB CA (controlled by the UL_CA_FAST_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter) requires that uplink intra-FR inter-band CA be enabled.

7.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Hyper Cell	NRDUCell.NrDuCellNetworkingMode	<i>Hyper Cell</i>	If the CA_FLEX_DATA_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter is deselected, downlink fast inter-gNodeB CA cannot be enabled in hyper cells, whose NRDUCell.NrDuCellNetworkingMode is set to HYPER_CELL .
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode	<i>Cell Combination</i>	Downlink fast inter-gNodeB CA cannot be enabled in combined cells if either of the following conditions is met: - The NRDUCellMultiTrp.DataTransMode parameter is set to BASIC_MODE. - The NRDUCellMultiTrp.DataTransMode parameter is set to DPS_MODE, and the CA_FLEX_DATA_SPLIT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoSwitch parameter is deselected.
Low-frequency TDD	Combined cell in joint transmission mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE and NRDUCellMultiTrp.DataTransMode set to JOINT_MODE	<i>Cell Combination</i>	Combined cells in joint transmission mode cannot work with downlink fast intra-gNodeB CA.

7.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Maximum-bandwidth reduction	OVERHEATING_ASSISTANCE_INFO_SW and UE_BW_REDUCE_SW options of the NRCellAlgoSwitch.UeAssistanceInfoSwitch parameter	<i>UE Power Saving</i>	When a UE requests maximum-bandwidth reduction, the base station deactivates all SCells of the UE in multi-carrier scenarios.
Low-frequency TDD	PDCCH skipping	PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter	<i>UE Power Saving</i>	With both downlink fast CA and PDCCH skipping enabled: - If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is deselected, PDCCH skipping does not take effect for UEs with SCells configured. - If the CA_UE_PDCCH_SKIP_SW option of the NRDUCellUePwrSaving.SchPwrSavingSw parameter is selected, PDCCH skipping can take effect for CA UEs with SCells not activated, but not for CA UEs with SCells activated.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSSG switching	UE_PWR_SAVING_ENHANCE_SW and SSSG_SWITCHING_SW options of the NRDUCellUePwrSaving.UePowerSavingSwitch parameter	<i>UE Power Saving</i>	With both downlink fast CA and SSSG switching enabled, a sparse SSSG can be configured for PCCs but not for SCCs. In this case, SCell activation for CA requires switching from a sparse SSSG to a dense SSSG and therefore is delayed.
Low-frequency TDD	CCE-based data split stop for CA UEs	CCE_BASED_CA_SPLIT_STOP_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	When CCE-based data split stop for CA UEs is enabled, it is recommended that downlink fast inter-gNodeB CA be also enabled to improve the flexibility of stopping data distribution to inter-gNodeB SCells.
Low-frequency TDD	Intra-base-station inter-carrier frequency selective scheduling	CROSS_CARRIER_FREQ_SEL_SCH_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	It is recommended that downlink fast intra-gNodeB CA be enabled together with intra-base-station inter-carrier frequency selective scheduling to prevent possible deterioration in the user experience of small-packet UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Multi-frequency joint scheduling	• MULTI_FREQ_CONV_ERGENT_SCH_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter • CA_CONCURRENT_SCH_CAPB_INCR_SW option of the gNodeBAlgo.DLSchSwitch parameter • CA_PRFB_BOOST_BSD_ON_MD_OPT_SW option of the NRDUCellCarrMgmt.CaEnhancedAlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	• Multi-frequency convergent scheduling not only enables data transmission to start simultaneously in SCells and the PCell, but also makes CA scheduling more precise. Therefore, when multi-frequency convergent scheduling is in effect, enabling downlink fast intra-gNodeB CA will not deliver gains. • When downlink fast intra-gNodeB CA is in effect, enabling multi-frequency convergent scheduling will only make CA scheduling more precise and will not provide the gains brought by simultaneous start of data transmission in SCells and the PCell. • It is recommended that downlink fast intra-gNodeB CA be enabled before CA Performance Boost Based on Multi-dimensional Optimization is

RAT	Function Name	Function Switch	Reference	Description
				enabled. If downlink fast intra-gNodeB CA is not enabled first, the gains of CA Performance Boost Based on Multi-dimensional Optimization may decrease.
Low-frequency TDD	Single DCI	DL_SINGLE_DCI_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	Downlink fast inter-gNodeB CA does not take effect for 3CC CA UEs for which single DCI has taken effect.

NOTE

Downlink fast CA requires downlink intra-band CA and downlink intra-FR inter-band CA. Apart from what are listed above, downlink fast CA shares the function impacts of downlink intra-band CA and downlink intra-FR inter-band CA as described in *Carrier Aggregation*.

7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

7.3.4 Networking

Downlink fast intra-gNodeB CA cannot be enabled in NR SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

7.3.5 Others

The other requirements of downlink fast intra-gNodeB CA and uplink fast intra-gNodeB CA are the same as those described in *Carrier Aggregation*.

The other requirements of downlink fast inter-gNodeB CA are the same as those described in [5.3.5 Others](#).

7.4 Operation and Maintenance

7.4.1 Data Configuration

7.4.1.1 Data Preparation

[Table 7-1](#) describes the parameters used for function activation. No parameters are involved in function optimization.

Table 7-1 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	CA_ULTRA_LO W_LATENCY_ SPLIT_SW	You are advised to select this option.
CA Enhanced Algorithm Extended Switch	NRDUCellCar rMgmt.CaEn hancedAlgoE xtSwitch	INTER_GNB_ ULTRALOW_L AT_SPLT_SW	You are advised to select this option.
CA Enhanced Algorithm Switch	NRDUCellCar rMgmt.CaEn hancedAlgoS witch	UL_CA_FAST_ SPLIT_SW	You are advised to select this option.

7.4.1.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling downlink fast intra-gNodeB CA for the PCell
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_ULTRA_LOW_LATENCY_SPLIT_SW-1;
//Enabling downlink fast inter-gNodeB CA for both the PCell and the SCell (using 2CC aggregation as an example)
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=INTER_GNB_ULTRALOW_LAT_SPLIT_SW-1;
MOD NRDUCELLCARRMGMT: NrDuCellId=1,
CaEnhancedAlgoExtSwitch=INTER_GNB_ULTRALOW_LAT_SPLIT_SW-1;
//Enabling uplink fast intra-gNodeB CA for the PCell
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_CA_FAST_SPLIT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling downlink fast inter-gNodeB CA for both the PCell and the SCell (using 2CC aggregation as an example)
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=INTER_GNB_ULTRALOW_LAT_SPLIT_SW-0;
MOD NRDUCELLCARRMGMT: NrDuCellId=1,
CaEnhancedAlgoExtSwitch=INTER_GNB_ULTRALOW_LAT_SPLIT_SW-0;
//Disabling downlink fast intra-gNodeB CA for the PCell
MOD NRDUCELLCARRMGMT: NrDuCellId=0,
CaEnhancedAlgoExtSwitch=CA_ULTRA_LOW_LATENCY_SPLIT_SW-0;
//Disabling uplink fast intra-gNodeB CA for the PCell
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaEnhancedAlgoSwitch=UL_CA_FAST_SPLIT_SW-0;
```

7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

7.4.2 Activation Verification

If the average throughput of CA SCell-activated UEs increases after downlink fast intra-gNodeB CA is enabled, downlink fast intra-gNodeB CA has taken effect. This average throughput is calculated using the following formula:

$$\frac{(N.CA.SccActiveUser.ThpVol.DL - N.CA.ThpVol.DL.LastSlot)/N.CA.ThpTime.DL.RmvLastSlot)}{}$$

If the average throughput of SCell-activated inter-gNodeB CA UEs increases after downlink fast inter-gNodeB CA is enabled, downlink fast inter-gNodeB CA has taken effect. This average throughput is calculated using the following formula:
$$\frac{(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot})}{N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}}$$

If the value of **N.CA.SCell.ThpVol.Ul.Cell** increases after uplink fast intra-gNodeB CA is enabled, uplink fast intra-gNodeB CA has taken effect. (This counter measures the total volume of uplink data received at the MAC layer from uplink CA UEs that treat the local cell as their SCell.)

7.4.3 Network Monitoring

The method of network monitoring for intra-band fast intra-gNodeB CA is the same as that described in *Carrier Aggregation*.

The method of network monitoring for intra-FR inter-band fast intra-gNodeB CA is the same as that described in *Carrier Aggregation*.

The method of network monitoring for downlink fast inter-gNodeB CA is the same as that described in [5.4.3 Network Monitoring](#), except that the average throughput of SCell-activated inter-gNodeB CA UEs is calculated using the following formula:
$$\frac{(N.\text{IntergNB.CA.SccActiveUser.ThpVol.DL} - N.\text{IntergNB.CA.ThpVol.DL.LastSlot})}{N.\text{IntergNB.CA.ThpTime.DL.RmvLastSlot}}$$

8 Massive CA

8.1 Principles

Basic Function of Massive CA

Massive CA aggregates four, five, intra-FR inter-band carriers in the downlink to provide higher bandwidth. With downlink intra-FR inter-band CA enabled (by selecting the **INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter), 4CC/5CC aggregation is enabled by setting the **NRDUCellCarrMgmt.CaDlMaxCcNum** parameter to **DL4CC/DL5CC**.

The CCs for massive CA are a set of intra-FR inter-band carriers. Not any two of them belong to the same frequency band or, when downlink intra-band CA is enabled (controlled by the **INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.CaAlgoSwitch** parameter), some of them may belong to the same frequency band.

Massive CA is supported only in low frequency bands. For the band+bandwidth combinations for this function, see section "CA Band+Bandwidth Combinations" in *Carrier Aggregation*.

NR TDD massive CA is available for trial use.

The following functions are supported: frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs, reselection policy for 3CC and massive CA UEs in idle mode, and downlink RLC throughput distribution measurement for 3CC or higher CA UEs. For details, see [6.1 Principles](#).

 NOTE

Trial functions are functions that are not yet ready for full commercial release for certain reasons. For example, the industry chain (terminals/CN) may not be sufficiently compatible. However, these functions can still be used for testing purposes or commercial network trials. Anyone who desires to use the trial functions shall contact Huawei and enter into a memorandum of understanding (MoU) with Huawei prior to an official application of such trial functions. Trial functions are not for sale in the current version but customers may try them for free.

Customers acknowledge and undertake that trial functions may have a certain degree of risk due to absence of commercial testing. Before using them, customers shall fully understand not only the expected benefits of such trial functions but also the possible impact they may exert on the network. In addition, customers acknowledge and undertake that since trial functions are free, Huawei is not liable for any trial function malfunctions or any losses incurred by using the trial functions. Huawei does not promise that problems with trial functions will be resolved in the current version. Huawei reserves the rights to convert trial functions into commercial functions in later R/C versions. If trial functions are converted into commercial functions in a later version, customers shall pay a licensing fee to obtain the relevant licenses prior to using the said commercial functions. If a customer fails to purchase such a license, the trial function(s) will be invalidated automatically when the product is upgraded.

8.2 Network Analysis

8.2.1 Benefits

Massive CA increases the downlink peak data rates of UEs.

Table 8-1 lists the ranges of peak data rates that a CA UE can achieve using downlink 4CC aggregation.

Table 8-1 Ranges of peak data rates achieved using downlink 4CC aggregation

SCell Weight Type	Peak Data Rate Range
SRS-based	(A x 98% + B x 98% + C x 98% + D x 95%) to (A x 100% + B x 100% + C x 100% + D x 100%)
PMI-based	(A x 98% + B x 95% + C x 95% + D x 95%) to (A x 100% + B x 100% + C x 100% + D x 100%)
Note 1: The SCell weight type depends on CA UE capabilities. If a CA UE transmits SRSs in an SCell, SRS-based weights are used. Otherwise, PMI-based weights are used. Note 2: A denotes the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. Note 3: B, C, or D denotes the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights.	

Table 8-2 lists the ranges of peak data rates that a CA UE can achieve using downlink 5CC aggregation.

Table 8-2 Ranges of peak data rates achieved using downlink 5CC aggregation

SCell Weight Type	Peak Data Rate Range
SRS-based	(A x 98% + B x 98% + C x 98% + D x 95% + E x 95%) to (A x 100% + B x 100% + C x 100% + D x 100% + E x 100%)
PMI-based	(A x 98% + B x 95% + C x 95% + D x 95% + E x 95%) to (A x 100% + B x 100% + C x 100% + D x 100% + E x 100%)
Note 1: The SCell weight type depends on CA UE capabilities. If a CA UE transmits SRSs in an SCell, SRS-based weights are used. Otherwise, PMI-based weights are used. Note 2: A denotes the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as the PCell for the UE. Note 3: B, C, D, or E denotes the downlink data rate that the UE in the single-carrier state can achieve in a cell that is later configured as an SCell and uses SRS- or PMI-based weights.	

 NOTE

In the case of 4CC/5CC aggregation, SRS-based weights can be used in at most two SCells.

In the case of 4CC/5CC aggregation, if the PCell of a UE uses the dual-period 8:2 slot configuration and an SCell uses the single-period 8:2 slot configuration or the other way round, one slot in the SCell is unavailable for scheduling due to the restriction that no more than eight k1 values can be configured for a single UE in section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331. This results in a 1/8 loss in the downlink peak rate of the UE in this SCell, without affecting the downlink peak rate in the PCell. It also affects the overall downlink peak rate of the UE. The amount of the impact varies depending on the PCell and SCell bandwidths. For example, if the two cells have the same bandwidth, there will be a 1/16 loss. The loss in the overall downlink peak rate of the UE is even greater when there are UE compatibility issues.

The compatibility with single-period 8:2 and dual-period 8:2 slot configuration combinations is controlled by the **CA_SLOT_CONFIG_EXT_SW** option of the **NRCellCaMgmtConfig.CaStrategySwitch** parameter. For details about the slot configurations, see the "Slot Configuration Combinations" section in *Carrier Aggregation*.

After the "reselection policy for 3CC and massive CA UEs in idle mode" function is enabled, the cell may have an increased proportion of 3CC and massive CA UEs with specific operator and QCI details, and there may be an increase in the average throughput of SCell-configured CA UEs ((**N.CA.SccConfigUser.ThpVol.DL** – **N.CA.SccConfigUser.ThpVol.DL.LastSlot**)/**N.CA.SccConfigUser.ThpTime.DL.RmvLastSlot**).

8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

The network impacts of massive CA are the same as those of downlink intra-FR inter-band CA and downlink intra-band CA. For details, see *Carrier Aggregation*.

If uplink enhancement for smart carrier selection takes effect in suitable scenarios, the number of UEs for which massive CA takes effect decreases. As a result, the

average number of downlink CA UEs that treat the local cell as their PCell (**N.User.CA.PCell.DL.Avg**) decreases. For details about uplink enhancement for smart carrier selection, see *Carrier Aggregation*.

8.3 Requirements

8.3.1 Licenses

Before enabling massive CA, apply for the licenses for both 3CC CA and Massive CA.

Feature ID	Feature Name	Model	Sales Unit
FOFD-080012	Massive CA	NR0S00MMCA00	per Cell
FOFD-051211	3CC CA	NR0SDL3CCA00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

8.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	- Downlink massive CA with some carriers in the same band and others in different bands of the same FR requires both downlink intra-FR inter-band CA and intra-band CA to be enabled. - Downlink massive CA with all carriers in different bands of the same FR requires downlink intra-FR inter-band CA to be enabled.
Low-frequency TDD	Intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	
Low-frequency TDD	Frequency-priority-based inter-frequency handover	FREQ_PRIORITY_BASED_HO option of the NRCCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs can take effect only when frequency-priority-based inter-frequency handover is in effect.
Low-frequency TDD	Frequency-priority-based inter-frequency PSCell change	FREQ_PRIORITY_BASED_HO option of the NRCCellAlgoSwitch.InterFreqHoSwitch parameter	<i>NSA Mobility Management</i>	Frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs can take effect only when frequency-priority-based inter-frequency PSCell change is in effect.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SRS interference coordination based on self-contained slots	SRS_INTRF_COORD_S_SLOT_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	<i>Channel Management</i>	The SRS_INTRF_COORD_S_SLOT_SW option of this parameter must be selected if (1) NRDUCellCarrMgmt.CaDLMaxCcNum is set to DL3CC/DL4CC/DL5CC and (2) the switch for SRS remote interference avoidance is turned on.
Low-frequency TDD	SRS remote interference avoidance	SRS_RIM_INTRF_AVOID_SW option of the NRDUCellSrsSrsAlgoExtSwitch parameter	None	

8.3.2.2 Mutually Exclusive Functions

The mutually exclusive functions for downlink massive CA are the same as [6.3.2.2 Mutually Exclusive Functions](#) for 3CC aggregation.

8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Independent SUL deployment	SUPER_UL_TDM_SCH_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Downlink massive CA does not take effect in independent SUL deployment mode.
Low-frequency TDD	SUL and NR FDD co-deployment	FLEX_SUPER_UPLINK_SW option of the NRDUCellAlgoSwitch.SuperUplinkSwitch parameter	<i>Super Uplink</i>	Downlink massive CA does not take effect in SUL and NR FDD co-deployment mode.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic-codebook-based HARQ feedback over the PUCCH	PUCCH_DYN_HARQ_ACK_CB_SW option of the NRDUCellCarrierMgmt.CaEnhancedAlgorithmExtSwitch parameter	<i>Carrier Aggregation</i>	Dynamic-codebook-based HARQ feedback over the PUCCH does not take effect for UEs in the massive CA state.

NOTE

Apart from what are listed above, downlink massive CA shares the function impacts of 3CC aggregation as described in [6.3.2.3 Function Impacts](#).

8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

NOTE

One prerequisite for CA is that the corresponding cells have been set up. The following requirements for base station models, boards, and RF modules are based on this prerequisite.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the related BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

CA can be performed in one baseband processing unit or between baseband processing units.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

8.3.4 Networking

Inter-gNodeB 5CC aggregation requires that the version of the base station that serves each SCell be V100R020C00 or later.

The other networking requirements of downlink intra-FR inter-band massive CA are the same as those described in *Carrier Aggregation*.

8.3.5 Others

The other requirements of downlink intra-FR inter-band massive CA are the same as those described in *Carrier Aggregation*.

8.4 Operation and Maintenance

8.4.1 Data Configuration

8.4.1.1 Data Preparation

Before activating this function, ensure that the **NRCellRelation**, **NRCellFreqRelation**, and **NRExternalNCell** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode* and *NSA Mobility Management*.

Table 8-3 describes the parameters used for function activation. The following parameters do not take effect for SUL cells, for which the **NRDUCell.DuplexMode** parameter is set to **CELL_SUL**.

Before enabling intra-FR inter-band massive CA, enable downlink intra-FR inter-band CA by preparing data as described in *Carrier Aggregation*.

Table 8-3 Parameters used for activation

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Band 1	gNBCustCaB andComb.FrequencyBand1	N/A	This parameter specifies CA band 1. Set this parameter as required.
Frequency Bandwidth 1	gNBCustCaB andComb.FrequencyBandwidth1	N/A	This parameter specifies the bandwidth of CA band 1. Set this parameter as required.
Frequency Band 2	gNBCustCaB andComb.FrequencyBand2	N/A	This parameter specifies CA band 2. Set this parameter as required.

Parameter Name	Parameter ID	Option	Setting Notes
Frequency Bandwidth 2	gNBCustCaB andComb.FrequencyBandwidth2	N/A	This parameter specifies the bandwidth of CA band 2. Set this parameter as required.
Frequency Band 3	gNBCustCaB andComb.FrequencyBand3	N/A	This parameter specifies CA band 3. Set this parameter as required.
Frequency Bandwidth 3	gNBCustCaB andComb.FrequencyBandwidth3	N/A	This parameter specifies the bandwidth of CA band 3. Set this parameter as required.
Frequency Band 4	gNBCustCaB andComb.FrequencyBand4	N/A	This parameter specifies CA band 4. Set this parameter as required.
Frequency Bandwidth 4	gNBCustCaB andComb.FrequencyBandwidth4	N/A	This parameter specifies the bandwidth of CA band 4. Set this parameter as required.
Frequency Band 5	gNBCustCaB andComb.FrequencyBand5	N/A	This parameter specifies CA band 5. Set this parameter as required.
Frequency Bandwidth 5	gNBCustCaB andComb.FrequencyBandwidth5	N/A	This parameter specifies the bandwidth of CA band 5. Set this parameter as required.
CA Downlink Max CC Number	NRDUCellCarrierMgmt.CaDlMaxCcNum	N/A	Downlink massive CA requires that this parameter be set to DL4CC/DL5CC .
Inter-Frequency Handover Switch	NRCelAlgoSwitch.InterFreqHoSwitch	3CC_FREQ_PRIORITY_BASED_HO	Select this option if frequency-priority-based inter-frequency handover (in SA networking) or frequency-priority-based inter-frequency PSCell change (in NSA networking) is expected to take effect only for 3CC UEs and massive CA UEs.
Service Differentiation Algorithm Switch	NRCelOpQciBearer.ServiceDiffAlgoSwitch	MASS_CA_IDLE_RESEL_POLICY_SW	It is recommended that this option be selected if the proportion of 3CC and massive CA UEs with specific operator and QCI details is expected to be increased.

Parameter Name	Parameter ID	Option	Setting Notes
UE-specific Statistics Policy	NRDUCellOamParam.UeSpecStatisticsPolicy	N/A	If the downlink RLC throughput distributions of 3CC or higher CA UEs are to be measured, set NRDUCellOamParam.UeSpecStatisticsPolicy to DL_NCC_UE .

8.4.1.2 Using MML Commands

Before using MML commands, refer to [8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the maximum number of downlink carriers to enable downlink 4CC aggregation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDiMaxCcNum=DL4CC;
//((Optional) Adding a customized CA band+bandwidth combination for downlink 4CC aggregation
ADD GNBCUSTCABANDCOMB: CustCaBandCombId=1, BandwidthCombSet=0, FrequencyBand1=N77,
FrequencyBandwidth1=CELL_BW_100M, FrequencyBand2=N77, FrequencyBandwidth2=CELL_BW_90M,
FrequencyBand3=N41, FrequencyBandwidth3=CELL_BW_100M, FrequencyBand4=N41,
FrequencyBandwidth4=CELL_BW_90M;
//Setting the maximum number of downlink carriers to enable downlink 5CC aggregation
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDiMaxCcNum=DL5CC;
//((Optional) Adding a customized CA band+bandwidth combination for downlink 5CC aggregation
ADD GNBCUSTCABANDCOMB: CustCaBandCombId=1, BandwidthCombSet=0, FrequencyBand1=N77,
FrequencyBandwidth1=CELL_BW_100M, FrequencyBand2=N77, FrequencyBandwidth2=CELL_BW_90M,
FrequencyBand3=N41, FrequencyBandwidth3=CELL_BW_100M, FrequencyBand4=N41,
FrequencyBandwidth4=CELL_BW_90M, FrequencyBand5=N66, FrequencyBandwidth5=CELL_BW_100M;
//Turning on 3CC_FREQ_PRIORITY_BASED_HO if frequency-priority-based inter-frequency handover (in SA
networking) or frequency-priority-based inter-frequency PSCell change (in NSA networking) is expected to
take effect only for 3CC UEs and massive CA UEs
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=3CC_FREQ_PRIORITY_BASED_HO-1;
//((Optional) Turning on MASS_CA_IDLE_RESEL_POL_SW if the proportion of 3CC and massive CA UEs with
specific operator and QCI details is expected to be increased
ADD NRCELLOPQCIBEARER: NrCellId=0, OperatorId=0, Qci=0,
ServiceDiffAlgoSwitch=MASS_CA_IDLE_RESEL_POL_SW-1;
//((Optional) Setting UeSpecStatisticsPolicy to DL_NCC_UE if the downlink RLC throughput distributions of
3CC or higher CA UEs are to be measured
MOD NRDUCELLLOAMPARAM: NrDuCellId=0, UeSpecStatisticsPolicy=DL_NCC_UE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off 3CC_FREQ_PRIORITY_BASED_HO  
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=3CC_FREQ_PRIORITY_BASED_HO-0;  
//Turning off MASS_CA_IDLE_RESEL_POL_SW  
MOD NRCELLOPQCIBEARER: NrCellId=0, OperatorId=0, Qci=0,  
ServiceDiffAlgoSwitch=MASS_CA_IDLE_RESEL_POL_SW-0;  
//Disabling downlink massive CA  
MOD NRDUCELLCARRMGMT: NrDuCellId=0, CaDlMaxCcNum=NOT_CONFIG;
```

8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

8.4.2 Activation Verification

Provided that CA UEs are available on the network, downlink massive CA has taken effect if any counter in [Table 8-4](#) returns a non-zero value. These counters are related to the downlink data volume and data transmission duration regarding massive CA.

Table 8-4 Counters related to massive CA

Counter ID	Counter Name
1911842637	N.Massive.CA.ThpVol.DL
1911842639	N.Massive.CA.ThpVol.DL.LastSlot
1911842636	N.Massive.CA.ThpTime.DL.RmvLastSlot

[Table 8-5](#) lists the counters used to observe the numbers of attempts and successful executions regarding SCell configuration, change, activation, deactivation, and removal.

Table 8-5 Counters related to SCell management

Counter ID	Counter Name
1911816777	N.CA.SCell.Add.Att
1911816778	N.CA.SCell.Add.Succ
1911816781	N.CA.SCell.Mod.Att
1911816782	N.CA.SCell.Mod.Succ
1911816896	N.CA.SCell.Act.Att
1911816897	N.CA.SCell.Act.Succ
1911816898	N.CA.SCell.Deact.Att

Counter ID	Counter Name
1911816899	N.CA.SCell.Deact.Succ
1911816779	N.CA.SCell.Rmv.Att
1911816780	N.CA.SCell.Rmv.Succ

8.4.3 Network Monitoring

Monitor the **N.Massive.CA.ThpVol.DL** counter to observe the RLC traffic volume to UEs in the massive CA state in a cell. This counter measures the total volume of downlink data sent at the RLC layer in a cell to SCell-activated CA UEs in the massive CA state.

Observe the average throughput of SCell-activated UEs in the massive CA state, which is calculated using the following formula: **(N.Massive.CA.ThpVol.DL – N.Massive.CA.ThpVol.DL.LastSlot)/N.Massive.CA.ThpTime.DL.RmvLastSlot**.

 NOTE

Optimized measurement of downlink CA throughput can be enabled to improve the accuracy of measuring the average throughput of SCell-activated UEs. For details, see the descriptions of this function in *Carrier Aggregation*.

As massive CA requires the basic CA functions of downlink intra-band CA and downlink intra-FR inter-band CA, the network monitoring method for these basic CA functions can also be used to monitor the relevant counters. For details about the methods, see section "Network Monitoring" under "Downlink Intra-Band CA" and "Downlink Intra-FR Inter-Band CA" in *Carrier Aggregation*.

9 CA-UE-specific Frequency-Priority-based Inter-Frequency Handover

9.1 Principles

When multiple frequency bands provide overlapping coverage on a network, some carriers on the network may not cover contiguous areas. To optimize service continuity for CA UEs, PCCs must be anchored on frequencies providing contiguous coverage. On frequencies providing discontinuous coverage, a certain amount of traffic must be carried. The existing function of frequency-priority-based inter-frequency handover does not meet these requirements. Therefore, CA-UE-specific frequency-priority-based inter-frequency handover is introduced. This function is controlled by the **CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter. This function is recommended when the target frequency after a handover can ensure equivalent or more favorable conditions for CA to take effect as the current frequency. The conditions for CA to take effect include but are not limited to CA-related configurations and other influencing factors of CA gains. For details, see the related descriptions in *Carrier Aggregation* and this document.

CA-UE-specific frequency-priority-based inter-frequency handover comes in parallel with basic frequency-priority-based inter-frequency handover and RFSP-specific frequency-priority-based inter-frequency handover. For an individual UE, which of these inter-frequency handover functions takes place is determined as follows:

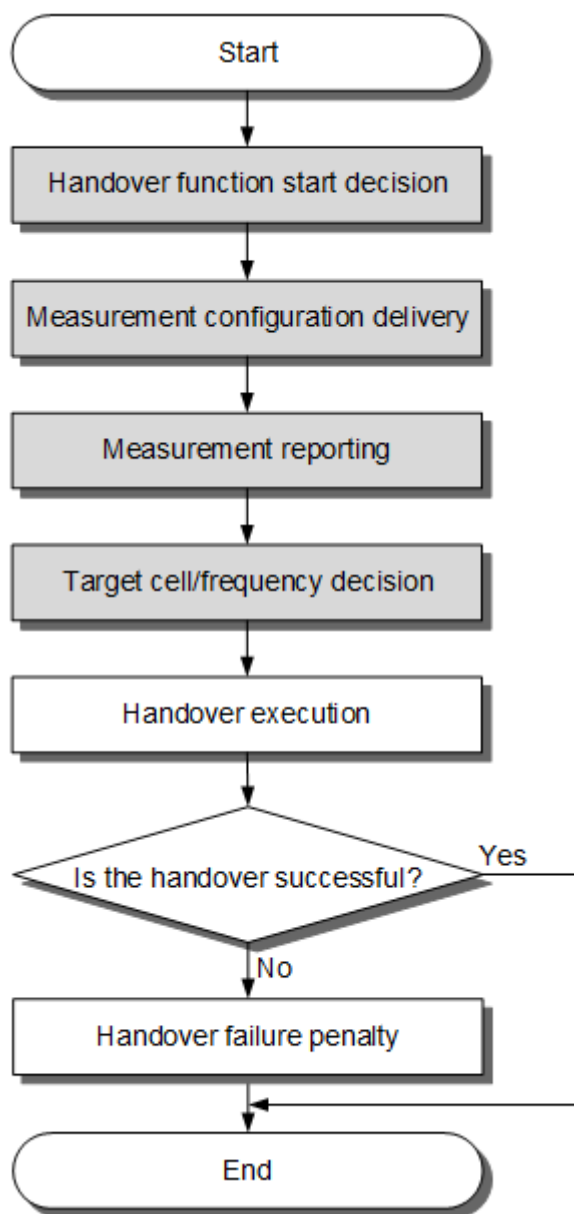
- An RFSP-specific frequency-priority-based inter-frequency handover takes place so long as the triggering conditions for this function are met.
- If the triggering conditions for RFSP-specific frequency-priority-based inter-frequency handover are not met:
 - A CA-UE-specific frequency-priority-based inter-frequency handover takes place so long as the triggering conditions for this function are met.
 - A basic frequency-priority-based inter-frequency handover takes place in other cases.

Of these functions:

- RFSP-specific frequency-priority-based inter-frequency handover, including its principles and triggering conditions, is described in chapter "Frequency-Priority-based Inter-Frequency Handover" in *SA Mobility Management in Connected Mode*.
- CA-UE-specific frequency-priority-based inter-frequency handover, including its principles and triggering conditions, is described in the present chapter.
- Basic frequency-priority-based inter-frequency handover is described as the function frequency-priority-based inter-frequency handover in *SA Mobility Management in Connected Mode* and the function frequency-priority-based inter-frequency PSCell change in *NSA Mobility Management*.

Figure 9-1 shows the overall procedure for a CA-UE-specific frequency-priority-based inter-frequency handover.

Figure 9-1 Overall procedure for a CA-UE-specific frequency-priority-based inter-frequency handover



In the figure, the elements in gray indicate the parts with differences from basic frequency-priority-based inter-frequency handover. These parts will be detailed in the present chapter. The elements in white indicate the parts that are the same as those in basic frequency-priority-based inter-frequency handover. For details about them, see *SA Mobility Management in Connected Mode*.

9.1.1 Handover Function Start Decision

CA-UE-specific frequency-priority-based inter-frequency handover takes effect only when all of the following conditions are met:

- The data configuration at the gNodeB must meet the following requirements:
 - The **CA_FREQ_PRI_BASED_INTERF_HO_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected for the PCell.
 - The basic CA switch, which refers to either the downlink intra-band CA switch or the downlink intra-FR inter-band CA switch, is turned on for the PCell. For details about the switches, see *Carrier Aggregation*.
- The UE capability must meet the requirement described in [Target CA Capability of UEs](#).

With the preceding conditions met, the gNodeB starts [9.1.2 Measurement Configuration Delivery](#) after receiving measurement reports indicating that the serving cell meets the entering condition for event A1 or A2. Which of event A1 or event A2 is used is determined by the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter.

- If this parameter is set to **A1_BASED**, the gNodeB delivers A1 measurement configurations. The phase of measurement configuration delivery starts when the gNodeB receives A1 measurement reports.
- If this parameter is set to **A2_BASED**, the gNodeB delivers A2 measurement configurations. The phase of measurement configuration delivery starts when the gNodeB receives A2 measurement reports.

NOTE

For the suggestions on whether to set **NRCellMobilityConfig.FreqPriHoTriggerMode** to **A1_BASED** or **A2_BASED**, see section "Triggering Mode of Frequency-Priority-based Handover" in *SA Mobility Management in Connected Mode*.

Events A1 and A2 are defined as follows:

- Event A1 indicates that the signal quality of the serving cell is higher than a threshold. Event A1 is triggered if its entering condition ($Ms - Hys > Thresh$) is true throughout *TimeToTrig*.
- Event A2 indicates that the signal quality of the serving cell is lower than a threshold. Event A2 is triggered if its entering condition ($Ms + Hys < Thresh$) is true throughout *TimeToTrig*.

In these descriptions:

- The definitions and values of *Ms* and *Hys* for events A1 and A2 are the same as those in basic frequency-priority-based inter-frequency handover. For details, see *SA Mobility Management in Connected Mode*.
- *Thresh* and *TimeToTrig* for events A1 and A2 can be set separately for CA-UE-specific frequency-priority-based inter-frequency handover.

- *Thresh* for event A1 = **NRCeIIInterFHoMeaGrp.FreqPriInterFA1RsrpThld** + **NRCeIIUserMeasCfg.RsrpThldOffset**, where the **NRCeIIInterFHoMeaGrp.FreqPriInterFA1RsrpThld** parameter specifies *Thresh* for event A1 in basic frequency-priority-based inter-frequency handover
- *Thresh* for event A2 = **NRCeIIInterFHoMeaGrp.FreqPriInterFA2RsrpThld** + **NRCeIIUserMeasCfg.RsrpThldOffset**, where the **NRCeIIInterFHoMeaGrp.FreqPriInterFA2RsrpThld** parameter specifies *Thresh* for event A2 in basic frequency-priority-based inter-frequency handover
- *TimeToTrig* for events A1 and A2 = **NRCeIIUserMeasCfg.TimeToTrig**
Before setting **NRCeIIUserMeasCfg.RsrpThldOffset** and **NRCeIIUserMeasCfg.TimeToTrig**, set **NRCeIIUserMeasCfg.UserType** to **CA_CAP_USER** and then **NRCeIIUserMeasCfg.AlgoEventType** to **FREQ_PRI_HO_A1A2**.

Target CA Capability of UEs

The gNodeB checks the UE-reported CA capabilities for the target CA capability. The target CA capability refers to either of the following:

- UE-reported CA capability which indicates that the operating frequency of the serving cell can act as the PCC.
- UE-reported CA capability which indicates that at least one target frequency with a non-zero priority can act as the PCC. The priority is specified by the **NRCeIIFreqRelation.CaAnchorFreqPriority** parameter for individual frequencies.

9.1.2 Measurement Configuration Delivery

The gNodeB selects frequencies and delivers related measurement configurations according to the following rules:

- Rules on frequency selection:
 - Neighboring frequencies with the **NRCeIIFreqRelation.CaAnchorFreqPriority** parameter set to **0** are not selected for delivery.
 - When only the basic CA switch is turned on for the serving cell, a neighboring frequency can be included in inter-frequency measurement configurations as long as its CA capability is higher than or equal to that of the serving frequency (that is, the neighboring frequency supports at least 2CC).
 - When 3CC aggregation is enabled by setting the **NRDUCellCarrMgmt.CaDIAMaxCcNum** parameter to **DL3CC** for the serving cell:
 - If the CA capability indicates that the serving frequency supports only single-carrier operation or 2CC aggregation, a neighboring frequency can be included in inter-frequency measurement configurations as long as its CA capability is higher than or equal to that of the serving frequency (that is, the neighboring frequency supports at least 2CC).

- If the CA capability indicates that the serving frequency supports 3CC aggregation or higher, a neighboring frequency can be included in inter-frequency measurement configurations as long as its CA capability is higher than or equal to that of the serving frequency. For example, if the serving frequency supports 3CC aggregation, a neighboring frequency can be included in inter-frequency measurement configurations as long as the neighboring frequency supports at least 3CC.

For details about 3CC aggregation, see [6 3CC Aggregation](#).

The gNodeB selects frequencies in descending order of the priority specified by the **NRCellFreqRelation.CaAnchorFreqPriority** parameter until the number of frequencies to be delivered reaches the maximum number of frequencies that can be delivered. For the maximum number, see section "Frequency Selection for Measurement" in "Basic Functions of Mobility Management in Connected Mode" in *SA Mobility Management in Connected Mode*. If multiple frequencies have the same priority, the NR frequency with a smaller **SsbFreqPos** value is preferentially selected.

- Rules on measurement configuration delivery:
 - If the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is set to **A1_BASED**, the gNodeB delivers measurement configurations related to events A5 and A2.
 - If the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is set to **A2_BASED**, the gNodeB delivers measurement configurations related to events A5 and A1.

NOTE

The event (A1 or A2) selected for measurement configuration is used to evaluate whether to stop the handover function and therefore is opposite to the event for evaluating whether to start this function. For example, if the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is set to **A1_BASED**, event A1 is used to evaluate whether to start the handover function. In this case, it is event A2 that is used to evaluate whether to stop this function.

Event A5 indicates that the signal quality of the serving cell becomes lower than threshold 1 and the signal quality of a neighboring cell becomes higher than threshold 2. Event A5 is triggered if its entering conditions ($Ms + Hys < Thresh1$) and ($Mn + Ofn + Ocn - Hys > Thresh2$) are true throughout *TimeToTrig*.

In these descriptions:

- The definitions and values of *Ms*, *Hys*, *Thresh1*, *Mn*, *Ofn*, and *Ocn* for event A5 are the same as those in basic frequency-priority-based inter-frequency handover. For details, see *SA Mobility Management in Connected Mode*.
- *Thresh2* and *TimeToTrig* for event A5 can be set separately for CA-UE-specific frequency-priority-based inter-frequency handover.
 - $Thresh2 = NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld + NRCellUserMeasCfg.RsrpThldOffset$, where the **NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld** parameter specifies *Thresh2* for event A5 in basic frequency-priority-based inter-frequency handover
 - $TimeToTrig = NRCellUserMeasCfg.TimeToTrig$

Before setting **NRCellUserMeasCfg.RsrpThldOffset** and **NRCellUserMeasCfg.TimeToTrig**, set **NRCellUserMeasCfg.UserType** to **CA_CAP_USER** and then **NRCellUserMeasCfg.AlgoEventType** to **FREQ_PRI_HO_A5**.

Whether the gNodeB delivers inter-frequency measurement configurations to the UE is subject to QCI-specific suppression of unnecessary inter-frequency handover and non-gap-assisted measurement. For details about QCI-specific suppression of unnecessary inter-frequency handover, see *SA Mobility Management in Connected Mode*. For details about non-gap-assisted measurement, see [Non-Gap-assisted Measurement](#).

Non-Gap-assisted Measurement

Non-gap-assisted measurement of target frequencies is controlled by the **CA_FREQ_PRI_BASED_HO_NO_GAP_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter.

- Option selected: If SCells have been configured for the UE, the gNodeB delivers A5 measurement configurations that contain only the operating frequencies of these SCells. According to 3GPP specifications, the UE will perform non-gap-assisted measurements on the operating frequencies of configured SCells. The gNodeB does not deliver measurement configurations in case that the operating frequencies of configured SCells do not overlap the set of frequencies to measure.
- Option deselected: The gNodeB delivers measurement configurations without considering whether configured SCells operate on the frequencies to measure.

9.1.3 Measurement Reporting

Measurement reporting for CA-UE-specific frequency-priority-based inter-frequency handover is the same as that for basic frequency-priority-based inter-frequency handover, except that event A4 is not involved. For details about this phase, see section "Measurement Reporting" in "Frequency-Priority-based Inter-Frequency Handover" of *SA Mobility Management in Connected Mode*.

9.1.4 Target Cell or Frequency Decision

Target cell or frequency decision for CA-UE-specific frequency-priority-based inter-frequency handover is the same as that for basic frequency-priority-based inter-frequency handover, except that event A4 is not involved. For details about this phase, see section "Target Cell or Frequency Decision" in "Frequency-Priority-based Inter-Frequency Handover" of *SA Mobility Management in Connected Mode*.

9.2 Network Analysis

9.2.1 Benefits

CA-UE-specific frequency-priority-based inter-frequency handover applies a separate frequency-priority-based handover policy to CA UEs to allow for flexible networking policies and meet operator requirements for service steering.

9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

This function has the following network impacts:

- This function increases the number of inter-frequency handovers and may change the inter-frequency handover success rate.
- If the **CA_FREQ_PRI_BASED_HO_NO_GAP_SW** option of the **NRCellAlgoSwitch.InterFreqHoSwitch** parameter is selected, better frequencies (if any) may not be selected, as A5 measurement configurations include only the operating frequencies of configured SCells in this case.
- If this function is enabled in a non-recommended scenario (see [9.1 Principles](#) for the recommended scenario), the values of CA-related indicators decrease, for example:
 - Indicators related to the number of CA UEs
 - Indicators related to the throughput of CA UEs

For details about other impacts, see the relevant "Impacts" section in *SA Mobility Management in Connected Mode*.

9.3 Requirements

9.3.1 Licenses

There are no license requirements.

9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	INTRA_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CA-UE-specific frequency-priority-based inter-frequency handover requires that either downlink intra-band CA or downlink intra-FR inter-band CA be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	INTRA_FR_INTER_BAND_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	CA-UE-specific frequency-priority-based inter-frequency handover requires that either downlink intra-band CA or downlink intra-FR inter-band CA be enabled.
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRIORITY_BASED_INTER_FREQ_HO_SW option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	The setting of the NRCellFreqRelation.CaAnchorFreqPriority parameter takes effect only if CA-UE-specific frequency-priority-based inter-frequency handover is enabled.
Low-frequency TDD	CA-UE-specific frequency-priority-based inter-frequency handover	CA_FREQ_PRIORITY_BASED_INTER_FREQ_HO_SW option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	Non-gap-assisted measurement (controlled by the CA_FREQ_PRIORITY_BASED_HO_NO_GAP_SW option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter) takes effect only when CA-UE-specific frequency-priority-based inter-frequency handover is enabled.
Low-frequency TDD	User Type	NRCellUserMeasCfg.UserType	<i>CA Experience Improvement</i>	If NRCellUserMeasCfg.AlgoEventType is set to FREQ_PRIORITY_HO_A1A2 or FREQ_PRIORITY_HO_A5 , NRCellUserMeasCfg.UserType can only be set to CA_CAP_USER .

9.3.2.2 Mutually Exclusive Functions

None

9.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and NRCellMlb.MlbTriggerMode set to CELL_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	Assume that both UE-number-based connected mode MLB and CA-UE-specific frequency-priority-based inter-frequency handover are enabled. If the target cell rejects a handover request for one of these functions, a handover cannot be initiated for the other function in the source cell within a short period of time.
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and NRCellMlb.MlbTriggerMode set to SPEC_EFF_BASED_USER_NUM	<i>Mobility Load Balancing in Connected Mode</i>	Assume that both spectral-efficiency-based connected mode MLB and CA-UE-specific frequency-priority-based inter-frequency handover are enabled. If the target cell rejects a handover request for one of these functions, a handover cannot be initiated for the other function in the source cell within a short period of time.
Low-frequency TDD	Radio-resource-based connected mode MLB	INTER_FREQ_CONNECTED_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter, and NRCellMlb.MlbTriggerMode set to RADIO_RESOURCE	<i>Mobility Load Balancing in Connected Mode</i>	Assume that both radio-resource-based connected mode MLB and CA-UE-specific frequency-priority-based inter-frequency handover are enabled. If the target cell rejects a handover request for one of these functions, a handover cannot be initiated for the other function in the source cell within a short period of time.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Latency-based optimal carrier selection	DELAY_BASED_CARR_SEL_SW option of the NRCellAlgoSwitch . MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	- After a UE is handed over to a cell for latency-based optimal carrier selection, CA-UE-specific frequency-priority-based inter-frequency handover does not take effect for the UE in this cell. - For a UE for which latency-based optimal carrier selection has not taken effect, CA-UE-specific frequency-priority-based inter-frequency handover takes effect when the relevant handover conditions are met.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch . <i>MultiFreqAlgoSwitch</i> parameter	<i>Multi-Frequency Smart Selection</i>	- When the smart carrier selection policy (specified by the NRCellCaMgmtConfig.<i>SmartCarrSelPolicy</i> parameter) is set to EXPERIENCE_FIRST, you are not advised to enable both experience-based smart carrier selection (controlled by the MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.<i>MultiFreqAlgoSwitch</i> parameter) and CA-UE-specific frequency-priority-based inter-frequency handover (controlled by the CA_FREQ_PRI_BASED_INTERF_HO_SW option of the NRCellAlgoSwitch.<i>InterFreqHoSwitch</i> parameter). If they are both enabled, ping-pong handovers may occur. - It is not recommended that event A2 be used for evaluation of CA-UE-specific frequency-priority-based inter-frequency handovers (that is, NRCellMobilityConfig.<i>FreqPriHoTriggerMode</i> is set to A2_BASED) when experience-based smart carrier selection is enabled. If event A2 is used, user experience of UEs at the cell edge or at a medium distance from the cell center may deteriorate.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coverage-based inter-frequency handover	COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	When event A2 is used for evaluation of CA-UE-specific frequency-priority-based inter-frequency handovers (that is, NRCellMobilityConfig.FreqPriHoTriggerMode is set to A2_BASED), it is recommended that the threshold for event A2 related to CA-UE-specific frequency-priority-based inter-frequency handovers (NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld + NRCellUserMeasCfg.RsrpThldOffset) be higher than the threshold for event A2 related to coverage-based inter-frequency handovers (NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld). If the thresholds do not meet this condition, event A2 cannot be used to trigger CA-UE-specific frequency-priority-based inter-frequency handovers.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	Assume that voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE and the FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell of the UE. During evaluation of a CA-UE-specific frequency-priority-based inter-frequency handover for the UE: • If the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set to TRUE for the target cell's operating frequency in the frequency group associated with the roaming UE and: – If the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a handover to the target cell. – If the roaming UE does not support handovers, the gNodeB determines that the CA-UE-specific frequency-priority-based inter-frequency handover is not performed for the roaming UE. • If the gNBFreqPriorityGroup.RoamingUeHplmnInterFreqHoFlag parameter is set

RAT	Function Name	Function Switch	Reference	Description
				to FALSE for the target cell's operating frequency in the frequency group associated with the roaming UE, the gNodeB determines that the CA-UE-specific frequency-priority-based inter-frequency handover is not performed for the roaming UE.
Low-frequency TDD	Energy-efficiency-based inter-frequency handover	ENERGY_EFFICIENCY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>Energy Conservation and Emission Reduction</i>	It is not recommended that CA-UE-specific frequency-priority-based inter-frequency handover and energy-efficiency-based inter-frequency handover be both enabled. If they are both enabled, energy saving gains will decrease.
Low-frequency TDD	Frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs	3CC_FREQ_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>CA Experience Improvement</i>	When both "frequency-priority-based inter-frequency handover dedicated to 3CC and massive CA UEs" and "CA-UE-specific frequency-priority-based inter-frequency handover" are enabled, only the latter takes effect.

9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

9.3.4 Networking

For details, see the "Networking" section of "Frequency-Priority-based Inter-Frequency Handover" in *SA Mobility Management in Connected Mode*.

9.4 Operation and Maintenance

9.4.1 Data Configuration

9.4.1.1 Data Preparation

Before activating this function, ensure that the **NRCellRelation** and **NRExternalNCell** MOs have been configured. For details about the configuration, see *SA Mobility Management in Connected Mode*.

For this function, set the parameters listed in [Table 9-1](#), in addition to the parameters described in the data preparation section of "Frequency-Priority-based Inter-Frequency Handover" in *SA Mobility Management in Connected Mode*.

NOTE

The relevant data preparation section in *SA Mobility Management in Connected Mode* may include a few parameters that are not required for CA-UE-specific frequency-priority-based inter-frequency handover, for example, the **CA_UE_FREQ_STEER_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter. Set parameters by referring to the principles for CA-UE-specific frequency-priority-based inter-frequency handover.

Table 9-1 Parameters dedicated to CA-UE-specific frequency-priority-based inter-frequency handover

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellFreqRelation.NrCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRCellFreqRelation.SsbFreqPos	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Frequency Band	NRCellFreqRelation.FrequencyBand	Set this parameter based on the network plan.
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	Select the CA_FREQ_PRI_BASED_INTERF_HO_SW option.
Inter-Frequency Handover Switch	NRCellAlgoSwitch.InterFreqHoSwitch	Select the CA_FREQ_PRI_BASED_HO_NO_GAP_SW option.
CA Anchor Frequency Priority	NRCellFreqRelation.CaAnchorFreqPriority	Set this parameter based on the network plan. The larger the value of this parameter, the higher the probability that CA UEs are handed over to cells on the frequency. The smaller the value of this parameter, the lower the probability.
User Type	NRCellUserMeasCfg.UserType	Set this parameter to CA_CAP_USER .
Algorithm Event Type	NRCellUserMeasCfg.AlgoEventType	- For events A1 and A2, set this parameter to FREQ_PRI_HO_A1 A2. - For event A5, set this parameter to FREQ_PRI_HO_A5.

Parameter Name	Parameter ID	Setting Notes
RSRP Threshold Offset	NRCellUserMeasCfg.RsrpThldOffset	Use the default value. If the NRCellUserMeasCfg.AlgoEventType parameter is set to FREQ_PRI_HO_A1A2, a larger value of this parameter results in a higher RSRP threshold for events A1 and A2 related to CA-UE-specific frequency-priority-based inter-frequency handover and a smaller value of this parameter results in a lower RSRP threshold. If the NRCellUserMeasCfg.AlgoEventType parameter is set to FREQ_PRI_HO_A5, a larger value of this parameter results in a higher RSRP threshold for event A5 related to CA-UE-specific frequency-priority-based inter-frequency handover and a smaller value of this parameter results in a lower RSRP threshold. This parameter only applies to threshold 2 for event A5.
Time to Trigger	NRCellUserMeasCfg.TimeToTrig	Use the default value. The smaller the value of this parameter, the higher the probability of reporting the corresponding event. The larger the value of this parameter, the lower the probability.

9.4.1.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

This section describes only the MML-based configuration dedicated to CA-UE-specific frequency-priority-based inter-frequency handover. For details about other MML-based configurations, see the relevant reference documents described in [9.4.1.1 Data Preparation](#).

Activation Command Examples

```
//Enabling CA-UE-specific frequency-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=CA_FREQ_PRI_BASED_INTERF_HO_SW-1;
//(TDD) Adding a frequency relationship for the NR cell, with the CA anchoring priority specified for the frequency
ADD NRCELLFREORELATION: NrCellId=0, SsbFreqPos=8000, FrequencyBand=N77, CaAnchorFreqPriority=1;
//(Optional) Setting parameters related to events A1, A2, and A5 for CA UEs
ADD NRCELLUSERMEASCFG: NrCellId=0, UserType=CA_CAP_USER, AlgoEventType=FREQ_PRI_HO_A1A2, RsrpThldOffset=0, TimeToTrig=320MS;
ADD NRCELLUSERMEASCFG: NrCellId=0, UserType=CA_CAP_USER, AlgoEventType=FREQ_PRI_HO_A5, RsrpThldOffset=0, TimeToTrig=320MS;
//(Optional) Enabling non-gap-assisted measurement for CA-UE-specific frequency-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=CA_FREQ_PRI_BASED_HO_NO_GAP_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling non-gap-assisted measurement for CA-UE-specific frequency-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=CA_FREQ_PRI_BASED_HO_NO_GAP_SW-0;
//Disabling CA-UE-specific frequency-priority-based inter-frequency handover
MOD NRCELLALGOSWITCH: NrCellId=0, InterFreqHoSwitch=CA_FREQ_PRI_BASED_INTERF_HO_SW-0;
```

9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

9.4.2 Activation Verification

Step 1 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**.

Step 2 In the navigation tree on the left of the **Signaling Trace Management** tab page, choose **NR > Application Layer > Uu Interface Trace**.

Step 3 Check the RRCReconfiguration message sent by the gNodeB. If the triggering condition for this function is met and this message indicates that the RSRP threshold for event A5 is equal to an existing threshold plus **NRCellUserMeasCfg.RsrpThldOffset**, CA-UE-specific frequency-priority-based inter-frequency handover has taken effect.

----End

9.4.3 Network Monitoring

This function can be monitored using the **Inter-Frequency Handover Out Success Rate (CU)**.

10 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

11 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

12 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

13 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.213: "NR; Physical layer procedures for control"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- *Carrier Aggregation*
- *UE Power Saving*
- *Channel Management*
- *Hyper Cell*
- *FWA*
- *Cell Combination*
- *Device-Pipe Synergy*
- *5G Networking and Signaling*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *NSA Mobility Management*
- *Energy Conservation and Emission Reduction*
- *Mobility Load Balancing in Connected Mode*
- *Mobility Load Balancing in Idle Mode*
- *Remote Interference Management (Low-Frequency TDD)*
- *NG and Xn Self-Management*
- *eXn Self-Management*
- *URLLC*
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- *Multi-Frequency Smart Selection*
- *Multi-Frequency Smart Aggregation*
- *Downlink Scheduling*
- *Transmission Resource Management*

- *Super Uplink*
- *Positioning*
- *License Control Item Lists*
- *VoNR*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *QoS Management*